Discover Internet of Things

RESEARCH Open Access

Adaptive lightweight intrusion detection system for internet of things and internet of medical things security

M. S. Padmini[1*], V. Amruth[2], A. S. Sindhu[3], M. Asha Rani[4], V. K. Annapurna[1] and C. S. Asha[5*]

*Correspondence:
M. S. Padmini
mspadmini19@nie.ac.in
C. S. Asha
asha.cs@manipal.edu
[1]Department of Computer Science and Engineering, The National Institute of Engineering (NIE), Mysuru, Karnataka, India
[2]Department of Information Science and Engineering, Maharaja Institute of Technology Mysore , Mysuru, Karnataka, India
[3]Department of Information Science and Engineering, JSS Science and Technology University (JSS STU), Mysuru, Karnataka, India
[4]Department of Computer Science and Engineering, GSSS Institute of Engineering and Technology for Women (GSSSIETW), Mysuru, Karnataka, India
[5]Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India

Abstract

The rapid development of Internet of Things (IoT) and Internet of Medical Things (IoMT) environments has increased the risks of cybersecurity, especially from DDoS, DoS, and other complex cyberattacks. Extensive research has been carried out on existing Machine Learning (ML) and Deep Learning (DL) based Intrusion Detection Systems (IDS); however, they still face several challenges such as class imbalance, high computational complexity, and poor generalization over heterogeneous datasets. This research addresses the gaps by proposing an Adaptive Weighted Feature-Optimized Lightweight Intrusion Detection System (AWFOL-IDS). The main goal is to design a scalable and efficient IDS that accurately detects attacks in both IoT and IoMT environments. AWFOL-IDS integrates Adaptive Weighted Information-Entropy Feature Selection (AWIE-FS) with a Weighted Adaptive Random Forest with Feature-Integrated Optimization (WARF-FIO) classifier with imbalance-aware sampling and weighted decision mechanisms. Experimental evaluation on CICIoT2023 and CICIoMT2024 datasets shows strong performance with 0.9989, 0.9968, and 0.9939 accuracy for 2-class, 8-class, and 34-class tasks and 0.9998, 0.9992, and 0.9856 accuracy for 2-class, 6-class, and 19-class tasks, respectively. Also, evaluation was conducted using 5-fold cross-validation, where findings showed better performance in terms of accuracy, precision, recall, and F-score. In conclusion, AWFOL-IDS is a lightweight and efficient solution for real-world intrusion detection in IoT and IoMT environments.

Keywords IoT security, Intrusion detection system, Machine learning, Class imbalance, Feature selection, IoMT

1 Introduction

The fast growth of Internet of Things (IoT) and Internet of Medical Things (IoMT) ecosystems has raised serious concerns over cybersecurity, as the attack surface for adversaries has expanded [1]. Distributed Denial of Service (DDoS), Denial of Service (DoS), spoofing, reconnaissance, and botnet-driven intrusions are taking over modern-day cyberattacks [2, 3]. In particular, DDoS attacks have become one of the most disruptive threats, and recent reports have shown a dramatic increase in both frequency

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

and intensity [4]. For example, over 20 million DDoS events were mitigated in a single quarter of 2025, and attack bandwidth reached unprecedented levels of over 30 Terabits per second (Tbps) in large-scale attacks [5]. Furthermore, cyber systems are attacked with billions of malicious attempts per day through automated botnets and compromised IoT devices [6, 7]. These numbers show that cyberattacks are not isolated events anymore but are instead continuous, large, and highly automated operations targeting critical infrastructures, such as healthcare, financial, and industrial systems [8, 9]. The increasing prevalence of IoT botnets, AI-assisted hacking tools, and multi-vector attack strategies has further intensified the severity of these threats, and traditional security mechanisms are no longer adequate [10].

The escalating scale and sophistication of cyberattacks targeting IoT and IoMT ecosystems motivate the need for Intrusion Detection Systems (IDS) that are simultaneously accurate, lightweight, and generalizable across heterogeneous environments. While existing ML- and DL-based IDS approaches report high detection accuracy, a closer examination of the literature reveals three persistent and interrelated gaps. First, most methods treat feature selection, class-imbalance handling, and classifier construction as independent, separately tuned stages [11, 12, 15, 17, 18], relying on static or dataset-agnostic feature subsets that cannot adapt when the relevance-redundancy trade-off shifts across tasks or datasets. Second, where class imbalance is addressed, it is typically handled through generic sampling or loss-reweighting strategies such as Synthetic Minority Over-Sampling Technique (SMOTE) applied uniformly across all features [14, 20, 23, 24], without linking the imbalance-correction mechanism to feature-level importance, so a minority-class signal carried by a weak, redundant feature is treated no differently than one carried by a highly discriminative feature. Third, techniques that do achieve strong accuracy through deep or hybrid architectures [16, 19, 20] or extensive hyperparameter tuning [16] incur high computational overhead, limiting their practicality for resource-constrained IoT and IoMT devices. Collectively, these gaps indicate the absence of an approach in which feature optimization and imbalance-aware learning are resolved jointly, rather than through disjoint pipelines, while remaining lightweight enough for deployment.

This motivates the design of Adaptive Weighted Feature-Optimized Lightweight Intrusion Detection System (AWFOL-IDS), which couples adaptive, entropy-based feature weighting directly with imbalance-aware classification to close this gap. The main goal of AWFOL-IDS is to design a scalable, efficient, and highly accurate IDS to work in general IoT as well as healthcare-specific IoMT environments. The AWFOL-IDS includes adaptive feature selection based on mutual information (MI) and redundancy analysis to identify the most discriminative features dynamically. It also employs a weighted RF classifier with imbalance-aware sampling, weighted decision tree construction, and adaptive ensemble aggregation. AWFOL-IDS also differs from the existing methods in that it deals explicitly with class imbalance, reduces feature redundancy, and keeps low computational complexity. Moreover, its adaptivity guarantees robustness on various datasets without dataset-specific tuning. AWFOL-IDS combines feature optimization and an enhanced lightweight ensemble learning framework to effectively mitigate the limitations of existing ML- and DL-based IDS techniques, while offering high detection accuracy and good generalization for IoT and IoMT security scenarios.

Unlike prior IDS studies that treat feature selection, class-imbalance handling, and ensemble construction as separate, independently tuned stages, the novelty of AWFOL-IDS lies in the adaptive coupling between these components. The relevance-redundancy trade-off used for feature weighting is not fixed by the user but is derived automatically from each dataset's class-variance and feature-variance statistics, and this same adaptive weight vector is propagated directly into the classifier's split criterion and imbalance-aware sampling stage, rather than being used only to filter features beforehand. This means feature importance and class-imbalance handling are resolved jointly, within a single adaptive mechanism, instead of through two independently configured pipelines. As a result, AWFOL-IDS requires no manual, dataset-specific parameter tuning to remain effective across heterogeneous IoT and IoMT traffic distributions, which distinguishes it from existing approaches that rely on static thresholds or fixed feature subsets regardless of the underlying data characteristics. The contributions of the work are as follows.

- Proposed an AWFOL-IDS for efficient intrusion detection in IoT and IoMT environments.
- Developed an Adaptive Weighted Information-Entropy Feature Selection (AWIE-FS) using MI and redundancy analysis to extract highly discriminative and non-redundant features.
- Introduced a dynamic weighting strategy that balances feature relevance and redundancy without manual parameter tuning.
- Designed a Weighted Adaptive Random Forest with Feature-Integrated Optimization (WARF-FIO) classifier incorporating class imbalance-aware sampling and weighted decision tree aggregation.
- Integrated imbalance handling techniques using class-based sample weighting and weighted bootstrap sampling to improve minority attack detection.
- Achieved a lightweight yet high-performance IDS architecture suitable for resource-constrained IoT devices.
- Demonstrated strong generalization capability through cross-dataset evaluation between CICIoT2023 and CICIoMT2024 datasets.
- Validated superior performance over traditional ML and DL approaches in terms of accuracy, precision, recall, and F1-score across binary and multi-class classification tasks.

Further, the manuscript is organized in the following manner. In Section II, the literature survey is discussed, where existing IDS approaches along with their limitations are discussed, providing a foundation for AWFOL-IDS. Further, Section III presents the AWFOL-IDS methodology in detail, with discussion of the results in Section IV and finally providing a conclusion for the work in Section V.

2 Literature survey

The recent studies on IoT security highlight the challenge of limited realistic datasets for effective attack detection and classification; hence, for addressing the following, C. Pinto et al. [11] aimed to develop a comprehensive dataset that reflected real-world IoT attack scenarios. This study designed a large-scale network of 105 IoT devices and systematically launched 33 attacks across seven classes, which included Mirai, spoofing,

brute force, web-based, reconnaissance, DoS, and DDoS, called the Canadian Institute for Cybersecurity Internet of Things 2023 (CICIoT2023). The work involved executing attacks from compromised IoT nodes targeting other devices to mimic realistic behavior. This dataset was evaluated using Deep Neural Network (DNN), RF, AB, Multi-Layer Perceptron (MLP), and LR. The testing was considered using 2 class, 7 class and 34 class where RF showed better performance achieving 99.68%, 99.43% and 99.16% respectively, for 80:20 training-testing ratio. The work did not handle class imbalance, as there are fewer samples in the benign class. Further, to address the lack of realistic datasets for Internet of Medical Things (IoMT) attack detection, S. Dadkhah et al. [12] focused on improving cybersecurity in healthcare environments where connected medical devices generate complex traffic patterns. Hence, developed a comprehensive dataset, called the Canadian Institute for Cybersecurity Internet of Medical Things 2024 (CICIoMT2024), which supported accurate attack classification using an ML approach. For developing the dataset, a testbed comprising 40 devices (15 simulated and 25 real devices) was constructed, where 18 attacks spanning spoofing, Message Queuing Telemetry Transport (MQTT), reconnaissance, DoS, and DDoS were executed using Bluetooth, MQTT, and Wireless Fidelity (Wi-Fi). The dataset included detailed device profiling and controlled data collection. The dataset was evaluated using RF, Deep Neural Network (DNN), AB, and LR, considering 19 classes, 6 classes, and 2 classes. The findings show that RF achieved better outcomes, achieving 73.3%, 73.5%, and 99.6%, respectively, for the training-testing ratio. The work did not consider handling class imbalance.

For improving IoT attack detection, M. A. Elaziz et al. [13] addressed the limitation of conventional DL approaches in achieving high accuracy with efficient parameter usage; hence, they aimed to design a novel intrusion detection approach using a Convolutional Kolmogorov-Arnold Network (CKAN) where traditional MLP layers in CNNs were replaced using Kolmogorov-Arnold Network (KAN) layers. The work involved training and evaluating CKAN considering the CICIoT2023, Network Security Laboratory - Knowledge Discovery in Databases (NSL-KDD), and ToN-IoT datasets. Along with the evaluation of CKAN, CNN, Recurrent Neural Network (RNN), and Auto-Encoder (AE) were also evaluated, where CKAN achieved better performance achieving better performance for all datasets for multi-class classification and binary classification. Specifically, for CICIoT2023, CKAN achieved 99.22% accuracy for 34 classes and 98.84% accuracy for 2 classes. The work did not consider handling the class imbalance issue. G. Akar et al. [14] focused on secure attack detection in IoMT environments, hence addressing the growing vulnerability of medical devices to sophisticated cyber threats. Hence, developed an effective IDS capable of accurately classifying multiple attack types using an enhanced LSTM method, which was designed for capturing temporal patterns in network traffic. The enhanced LSTM was evaluated using the CICIoMT2024 dataset for 2-class, 6-class, and 19-class, where LSTM achieved 100%, 98%, and 95% accuracy, respectively. For handling class imbalance, weight balancing was considered, i.e., by assigning more importance to the minority class during training to reduce imbalance. Also, used SMOTE, which generated synthetic samples for the minority class for handling class imbalance issue. Further, K. S. Adewole et al. [15] evaluated ensemble learning approaches, i.e., Categorical Boosting (CB), Light Gradient Boosting (LGBM), Extreme Gradient Boosting (XGB), AB, and RF on CICIoT2023 and Canadian Institute for Cybersecurity Intrusion Detection System 2017 (CIC-IDS2017), where XGB achieved 98.54%

and 99.91% accuracy for 2-class classification on a 70:30 training-testing ratio. The work failed to handle the class imbalance issue.

For addressing challenge of scalable and efficient IoT attack detection, M. V. C. Aragão et al. [16] presented a multi-stage ML approach which was aimed at achieving higher accuracy and reducing computational cost. The objective of the work was to develop a lightweight yet effective detection approach suitable for large-scale environments. The work integrated feature selection, data balancing, sampling, and hyperparameter tuning using XGB, LGBM, and a hybrid model, XGBRF. The work was evaluated using CICIoT2023 dataset, where XGB showed better performance for 2 class, 8 class and 34 class achieving 99.7%, 99.5%, and 99.4% accuracy respectively for 60:20:20 training, validation and testing ratio. In this work, class imbalance was addressed by using Hyperparameter Optimization (HPO) with random over-sampling and under-sampling, where the parameters were optimized using class-distribution thresholds using Tree-Structure Parzen Estimator (TPE) sampler in Optuna. J. Doménech et al. [17] addressed the limitation of using generic IoT datasets for IoMT attack detection, which leads to poor generalization in healthcare environments. Hence, evaluated the impact of dataset specificity on IDS performance and improved model reliability. The work involved training ML approaches on CICIoMT2024 and CICIoT2023 datasets, which was followed by cross-evaluation. Findings show that RF achieved 99.85% accuracy for 6-class classification when tested on CICIoMT2024 for an 80:20 training-testing ratio. Also, the findings show better performance when trained using the CICIoT2023 and tested using the CICIoMT2024 dataset. Further, the work did not handle the class imbalance issue, yet suggested using SMOTE for handling class imbalance. Further, A. Hafid et al. [18] implemented a fine-tuned XGB and compared it with LR, followed by a late fusion approach using max-voting for improving performance. The work was evaluated using the CICIoMT2024 dataset for 2-class classification, where fine-tuned XGB achieved 97% accuracy. The work did not handle class imbalance, but suggested using SMOTE for handling class imbalance.

I. A. Fares et al. [19] addressed the challenge of accurate IoT attack detection with limited data and high computational demands; hence, presented a hybrid transfer-learning-based IDS approach. The objective of the work was to improve detection performance while decreasing dependency on large-scale training data. The work combined Swin-Transformer (ST) for hierarchical feature extraction with LSTM for capturing temporal dependencies, which was followed by fine-tuning using transferred pre-trained weights. The ST-LSTM was evaluated using various datasets, including CICIoT2023, where it achieved 98.78% and 97.44% accuracy for 2 classes and 34 classes, respectively. The work did not consider the class imbalance issue. The work has also considered evaluation by training on CICIoT2023 and testing it on other datasets. M. Z. Khan et al. [20] aimed at designing a robust anomaly detection approach capable of identifying complex attack patterns in network traffic. Hence, the work integrated a Graph Convolutional Network (GCN) for capturing structural relationships and a transformer approach for learning sequential dependencies. The work was evaluated on the CICIoMT2024 dataset for 2-class, 6-class, and 19-class, where Transformer-GCN achieved 98.3%, 97.9%, and 98.28% accuracy, respectively, for an 80:20 training-testing ratio. In this work, SMOTE was used for handling class imbalance. J. C. García-Merino et al. [21] addressed the challenge of deploying an accurate lightweight attack detection

approach in resource-constrained IoT environments. The objective was to develop an efficient IDS by reducing feature dimensionality without compromising performance. The work involved applying various feature selection methods, i.e., Genetic Algorithm (GA), Principal Component Analysis (PCA), Permutation Feature Importance (PFI), MI, and $\mathcal{X}^2$ statistical tests to train the RF approach. The work was evaluated using an 80:20 training-testing ratio, using the CICIoT2023 dataset, where using RF with GA achieved 99.13% accuracy for 2-class classification. This work used an under-sampling approach for handling class imbalance.

M. Mahbub et al. [22] addressed the challenge of limited domain-specific datasets for accurate IoMT attack detection by exploring domain adaptation techniques; hence, their objective was to improve cross-domain intrusion classification using knowledge transferred from general IoT data. The work applied feature-based domain adaptation by introducing the Class-Wise Wasserstein Distance method for aligning source and target data distributions, combined with LR and Particle Swarm Optimization (PSO) for improved performance. The ML models considered included k-Nearest Neighbor (k-NN), MLP, DT, RF, and LR. The work was evaluated using different datasets, which also included the CICIoMT2024 dataset, only for 6-class classification. For the evaluation on the CICIoMT2024 dataset, RF achieved 97.51% accuracy. The work did not consider handling the class imbalance issue. R. Alfageer et al. [23] addressed the limitations of conventional IoMT IDS in handling unknown attacks and class imbalance; hence, they designed a robust approach capable of accurate multi-stage attack detection with open-set recognition. The work combined an AE for anomaly screening with supervised models, i.e., RF, XGB, and Lightweight CNN (LWCNN) for hierarchical classification, along with feature selection and various sampling configurations (oversampling and under-sampling). The work was evaluated using the CICIoMT2024 dataset, where 99.63% accuracy was achieved for 2-class classification using LWCNN with feature selection and oversampling. Further, for multi-class classification, RF with feature selection and oversampling achieved 99.77% accuracy. S. Palaniappan et al. [24] addressed the challenge of high-dimensional feature spaces in IoMT attack detection, which can hinder accuracy and efficiency in resource-limited environments. Hence, the authors aimed to develop an optimized feature selection approach that balanced detection performance and computation cost. The work combined MI filtering with a DL-based binary dragonfly algorithm, supported by a lightweight proxy network for reducing retraining overhead. The work was evaluated on the CICIoMT2024 dataset and further validated on IoMT-traffic data, where results showed up to 65% feature reduction while maintaining over 98% detection performance and significantly reducing processing time compared to conventional methods. The complete summary of the literature survey is given in Table 1.

The existing literature on IoT and IoMT intrusion detection reveals several limitations across different studies. The dataset development works in [11] and [12] achieve large-scale and realistic environments but fail to address the severe class imbalance problem, leading to biased learning toward majority attack classes. DL-based approaches like CKAN in [13] and ensemble methods in [15] provide high accuracy but lack explicit imbalance handling and may suffer from limited robustness. The enhanced LSTM model in [14] improves minority class detection using SMOTE and weighting; however, it introduces high computational complexity, making it less suitable for resource-constrained

Table 1 Literature survey summary

Ref	Method	Feature selection	Classifier	Handling class imbalance	Dataset	Results (Accuracy)	Train-valid-test ratio	Limitations
[11]	Large-scale IoT dataset development (CICIoT2023)	No	RF, DNN, MLP, AB, LR	No	CICIoT2023	99.68 (2 C), 99.43 (8 C), 99.16 (34 C)	80:00:20	Class imbalance not handled
[12]	IoMT dataset development (CICIoMT2024)	No	RF, DNN, AB, LR	No	CICIoMT2024	99.6 (2 C), 73.5 (6 C/19 C)	80:00:20	Class imbalance not handled
[13]	CKAN (CNN + KAN)	No	CKAN, CNN, RNN, AE	No	CICIoT2023, NSL-KDD, ToN-IoT	99.22 (34 C), 98.84 (2 C)	Not specified	No imbalance handling
[14]	Enhanced LSTM	No	LSTM	Yes (SMOTE, weights)	CICIoMT2024	100 (2 C), 98 (6 C), 95 (19 C)	Not specified	High computational cost
[15]	Ensemble learning	No	XGB, LGBM, RF, AB	No	CICIoT2023, CIC-IDS2017	99.91 (2 C)	70:0:30	No imbalance handling
[16]	Multi-stage ML (FS + tuning)	Yes	XGB, LGBM, XGBRF	Yes	CICIoT2023	99.7 (2 C), 99.5 (8 C), 99.4 (34 C)	60:20:20	Increased complexity
[17]	Cross-dataset evaluation	No	RF	No	CICIoMT2024, CICIoT2023	99.85 (6 C)	80:00:20	No imbalance handling
[18]	Fine-tuned XGB + fusion	No	XGB, LR	No	CICIoMT2024	97 (2 C)	Not specified	No imbalance handling
[19]	Swin Transformer + LSTM	No	ST-LSTM	No	CICIoT2023	98.78 (2 C), 97.44 (34 C)	Not specified	No imbalance handling
[20]	Transformer + GCN	No	Transformer-GCN	Yes (SMOTE)	CICIoMT2024	98.3 (2 C), 97.9 (6 C), 98.28 (19 C)	80:0:20	High complexity
[21]	Feature selection techniques	Yes	RF	Yes (under-sampling)	CICIoT2023	99.13 (2 C)	80:00:20	Possible data loss
[22]	Domain adaptation (Wasserstein)	No	RF, LR, MLP, k-NN	No	CICIoMT2024	97.51 (6 C)	Not specified	No imbalance handling
[23]	Multi-stage IDS (AE + ML)	Yes	RF, XGB, LWCNN	Yes (over/under-sampling)	CICIoMT2024	99.63 (2 C), 99.95 (5 C), 99.77 (16 C)	90:00:10	Complex architecture
[24]	Optimized feature selection (MI + Dragonfly)	Yes	ML/DL hybrid	Yes (Wrapper optimization)	CICIoMT2024	> 98	70:15:15	Not clearly addressing imbalance

IoT systems. Multi-stage ML models in [16] achieve strong accuracy but increase system complexity due to extensive feature selection and hyperparameter tuning. Cross-dataset evaluation in [17] and fine-tuned XGB in [18] do not effectively address imbalance, limiting their generalization reliability. Hybrid DL models such as ST-LSTM in [19] and Transformer-GCN in [20] achieve strong performance but suffer from high computational cost and dependence on oversampling techniques. Feature selection-based methods in [21] and [24] either risk information loss or do not explicitly resolve imbalance, while domain adaptation in [22] improves transferability but still lacks imbalance-aware learning. Multi-stage IDS in [23] improves performance using sampling strategies but results in complex architectures that are difficult to deploy. To overcome these limitations, AWFOL-IDS introduces an adaptive, lightweight, and imbalance-aware approach. It integrates MI-based adaptive feature selection with redundancy control, ensuring optimal and non-redundant feature subsets. The WARF-FIO further incorporates class-weighted sampling, weighted Gini splitting, and performance-based ensemble aggregation to effectively handle severe class imbalance while maintaining computational efficiency. Unlike existing methods, AWFOL-IDS achieves high accuracy, strong generalization across CICIoT2023 and CICIoMT2024, and low computational overhead, making it suitable for IoT and IoMT security applications.

Nevertheless, beyond the empirical gaps identified above, the closest related approaches differ from AWFOL-IDS in a structural respect that motivates the expected improvement in this work. In [11, 12, 15, 17], and [18], feature usage is either left unconstrained or reduced using static, dataset-independent criteria, so the same feature subset is applied regardless of how the relevance-redundancy balance shifts across tasks or datasets; because these methods do not adjust which features are trusted at split time, they are structurally unable to suppress a feature that is informative in one class distribution but redundant in another. In [14, 20, 23, 24], class imbalance is addressed through sampling or loss reweighting applied uniformly across all features, without linking the imbalance-handling mechanism to feature-level importance, so a minority-class signal carried by a low-weight, redundant feature receives the same treatment as one carried by a highly discriminative feature. AWFOL-IDS differs by propagating a single adaptive feature-weight vector (Eq. 4) into both the split criterion (Eq. 11) and, indirectly, the trees that receive imbalance-aware sampling (Eqs. 12–14), so that feature importance and imbalance handling are resolved jointly rather than independently. This coupling is the mechanism expected to yield the generalization and minority-class advantages evaluated in Sect. 4, rather than either component alone.

3 Methodology

To address limitations identified in existing approaches, as discussed in the literature survey, this work presents a novel lightweight and efficient approach called AWFOL-IDS. The AWFOL-IDS is specifically designed for generalized IoT environments, where the focus is on reducing computational complexity, dynamically selecting discriminative features according to attack categories, and handling class imbalance effectively. The architecture of AWFOL-IDS is presented in Fig. 1.

This section first discusses the datasets used in this study for training and testing AWFOL-IDS, followed by preprocessing of the datasets so that it can be trained in a generalizable way. Further, the feature selection process and classifier used in AWFOL-IDS

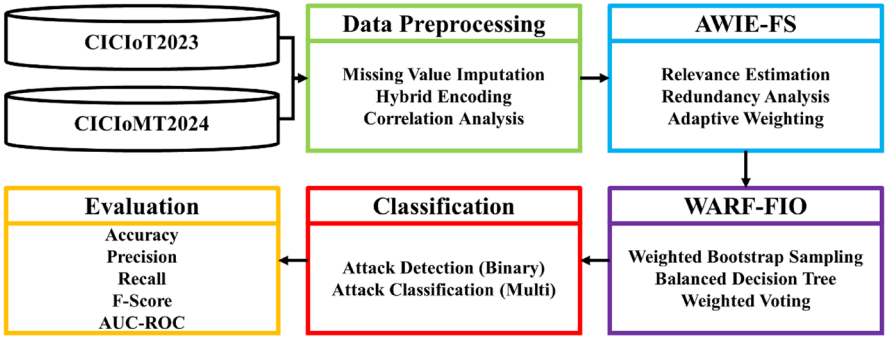


Fig. 1 AWFOL-IDS architecture

Table 2 Binary class distribution of CICIoT2023 dataset

Class	Samples
Benign	1,098,195
Attack	45,588,384
Total	46,686,579

are discussed in detail. In the final section, the complete process of AWFOL-IDS is presented in an algorithm.

3.1 Datasets

For this study, two realistic datasets, i.e., CICIoT2023 [25] and CICIoMT2024 [26], were used for evaluating the performance of IDS in both general and healthcare-specific IoMT environments. These datasets were designed to reflect real-world network behavior by incorporating various devices, protocols, and attack scenarios [11, 12]. The CICIoT2023 [11, 25] dataset represents a large-scale IoT network containing a total of 46,686,579 samples, as presented in Table 2. The dataset is highly imbalanced, having 45,588,384 attack samples and only 1,098,195 benign samples, which shows the challenge of detecting normal traffic accurately. Further, Table 3 presents the detailed distribution of attack categories and classes. The dataset includes various attack categories, in which DDoS attacks dominate, dataset having various subtypes.

Moreover, CICIoMT2024 [12, 26] focuses on the IoMT environment and contains 8,775,013 samples, as presented in Table 4. Similar to CICIoT2023, CICIoMT2024 also shows class imbalance, with 8,775,013 attack samples and 230,339 benign samples, as presented in Table 4. Further, Table 5 shows the distribution of attack categories. Compared to CICIoT2023, this dataset includes fewer samples, but provides more domain-specific attack scenarios with respect to medical devices and communication protocols.

Further, as presented in Table 6, the CICIoT2023 dataset includes a large and diverse set of 105 IoT devices, showing general-purpose environments like smart homes and Industrial IoT (IIoT). When compared with CICIoMT2024, it focuses on healthcare environments and has 40 devices, which include both real and simulated medical devices. While CICIoT2023 provides scale and diversity, CICIoMT2024 provides domain-specific insights, mainly for medical communication protocols like MQTT. This difference makes CICIoMT2024 more suitable for healthcare security work, whereas CICIoT2023 is better for general IoT intrusion detection.

Table 3 Category and class distribution of the CICIoT2023 dataset

Category	Class	Samples
Benign	Benign	1,098,195
DDoS	DDoS-ACK-Fragmentation	285,104
	DDoS-HTTP-Flood	28,790
	DDoS-ICMP-Flood	7,200,504
	DDoS-ICMP-Fragmentation	452,489
	DDoS-PSHACK-Flood	4,094,755
	DDoS-RSTFIN-Flood	4,045,285
	DDoS-Slow Loris	23,426
	DDoS-SYN-Flood	4,059,190
	DDoS-Synonymous IP-Flood	3,598,138
	DDoS-TCP-Flood	4,497,667
	DDoS-UDP-Flood	5,412,287
	DDoS-UDP-Fragmentation	286,925
DoS	DoS-SYN-Flood	2,028,834
	DoS-HTTP-Flood	71,864
	DoS-TCP-Flood	2,671,445
	DoS-UDP-Flood	3,318,595
Reconnaissance	Host Discovery	134,378
	OS Scan	98,259
	Port Scan	82,284
	Vulnerability Scan	37,382
	Ping Sweep	2262
Web-based	Backdoor-Malware	3218
	Browser Hijacking	5859
	Command Injection	5409
	SQL Injection	5245
	Uploading Attack	1252
	XSS	3846
Brute Force	Dictionary Brute Force	13,064
Spoofing	ARP-Spoofing	307,593
	DNS-Spoofing	178,911
Mirai	Mirai-Greeth Flood	991,866
	Mirai-GREIP Flood	751,682
	Mirai-UPD Plain	890,576

Table 4 Binary class distribution of the CICIoMT2024 dataset

Class	Samples
Benign	230,339
Attack	8,544,674
Total	8,775,013

Table 7 presents a detailed comparison of attack types in both datasets. The CICIoT2023 dataset includes 33 attack classes, which cover a wide range of categories. This provides more diversity, which includes network layer and application-layer attacks. When compared with the CICIoMT2024 dataset, which consists of 19 attack classes, having fewer categories and focuses more on healthcare-specific communication. While CICIoMT2024 shares common attack types with CICIoT2023, it introduces MQTT-based attacks, which are relevant in IoMT environments. However, it does not include web-based, brute-force, or Mirai attacks.

Table 5 Category and class distribution of the CICIoMT2024 dataset

Category	Class	Samples
Benign	Benign	230,339
DDoS	DDoS-SYN Flood	974,359
	DDoS-TCP Flood	987,063
	DDoS-ICMP Flood	1,887,175
	DDoS-UDP Flood	1,998,026
DoS	DoS-SYN Flood	540,498
	DoS-TCP Flood	462,480
	DoS-ICMP Flood	514,724
	DoS-UDP Flood	704,503
Reconnaissance	Ping Sweep	926
	Vulnerability Scan	3207
	OS Scan	20,666
	Port Scan	106,603
Spoofing	ARP Spoofing	17,791
MQTT	Malformed Data	6877
	DoS Connect Flood	15,904
	DDoS Connect Flood	214,952
	DoS Publish Flood	52,881
	DDoS Publish Flood	36,039

Table 6 Comparison of Devices in CICIoT2023 vs. CICIoMT2024

Feature	CICIoT2023	CICIoMT2024
Number of Devices	105 IoT devices	40 devices
Device Type	General IoT devices (cameras, sensors, routers, etc.)	Medical + simulated IoMT devices
Real Devices	Mostly real devices	25 real + 15 simulated
Application Domain	Smart home/general IoT	Healthcare/medical environment
Network Scale	Large-scale	Medium-scale
Protocols Used	Standard IoT protocols	Wi-Fi, Bluetooth, MQTT

Table 7 Comparison of attacks in CICIoT2023 vs. CICIoMT2024

Category	CICIoT2023 (33 Attacks)	CICIoMT2024 (19 Attacks)
DDoS	DDoS-ACK Fragmentation, DDoS-HTTP Flood, DDoS-ICMP Flood, DDoS-ICMP Fragmentation, DDoS-PSHACK Flood, DDoS-RSTFIN Flood, DDoS-Slow Loris, DDoS-SYN Flood, DDoS-Synonymous IP Flood, DDoS-TCP Flood, DDoS-UDP Flood, DDoS-UDP Fragmentation	DDoS-SYN Flood, DDoS-TCP Flood, DDoS-ICMP Flood, DDoS-UDP Flood
DoS	DoS-SYN Flood, DoS-HTTP Flood, DoS-TCP Flood, DoS-UDP Flood	DoS-SYN Flood, DoS-TCP Flood, DoS-ICMP Flood, DoS-UDP Flood
Reconnaissance	Host Discovery, OS Scan, Port Scan, Vulnerability Scan, Ping Sweep	Ping Sweep, Vulnerability Scan, OS Scan, Port Scan
Web-based	Backdoor Malware, Browser Hijacking, Command Injection, SQL Injection, Uploading Attack, XSS	-
Brute Force	Dictionary Brute Force	-
Spoofing	ARP Spoofing, DNS Spoofing	ARP Spoofing
Mirai	Mirai-GREETH Flood, Mirai-GREIP Flood, Mirai-UDP Plain	-
MQTT (IoMT-specific)	-	Malformed Data, DoS Connect Flood, DDoS Connect Flood, DoS Publish Flood, DDoS Publish Flood

3.2 Preprocessing

In this work, the preprocessing stage plays an important role in improving the quality and consistency of input data. The CICIoT2023 and CICIoMT2024 datasets have missing values and categorical variables, which can negatively impact performance. Hence, in this work, initially, missing and null values were handled. The numerical features having missing values were imputed using median-based imputation, which is more robust to outliers in comparison with mean imputation. For categorical features, missing values were replaced using the most frequent category (mode) to preserve distribution consistency. Further, the dataset contains several categorical attributes, such as protocol type or service identifiers. These were transformed into numerical representations using hybrid encoding, where low-cardinality features were encoded using one-hot encoding, while high-cardinality features were transformed using frequency encoding to reduce dimensionality. Further, highly correlated features were identified using correlation analysis, and only one representative features from highly correlated pairs were retained to reduce multicollinearity. Finally, the dataset was shuffled and split into training-testing sets (80:20, as seen in most of the studies) using a stratified method for preserving the original class distribution. Using the following approach ensured that the data were consistent for training and testing. To prevent data leakage between training and testing partitions, the train-test split was performed prior to any statistics-dependent step. Median values for numerical imputation, modal categories for categorical imputation, one-hot/frequency encoding mappings, and MI-based relevance and redundancy statistics used in Eqs. (1)-(3) were computed exclusively on the training partition; the fitted transformations and the resulting AWIE-FS feature subset were then applied to the test partition without re-computation. Stratified splitting preserved class proportions across partitions while avoiding duplication of individual flow records between splits.

3.3 Feature selection

For reducing dimensionality while preserving discriminative power, this work presents the AWIE-FS approach. The AWIE-FS method dynamically assigns importance to features by considering both relevance to the target variable and its redundancy with other features. Let the dataset be represented as $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{\mathcal{N}}$, where x_i represents the feature vector and $y_i \in \{0,1,\ldots,K\}$ representing the class label. The relevance of each feature f_j with respect to the class label is computed using MI using Eq. (1).

$$\mathcal{R}(f_j) = \mathcal{I}(f_j; y) = \sum_{f_j, y} p(f_j, y) \left(\log \frac{p(f_j, y)}{p(f_j)\, p(y)} \right) \tag{1}$$

Equation (1) measures how much information the features contribute to predicting the class label. In high-dimensional IoT and IoMT datasets, multiple features often capture overlapping or the same information, which leads to redundancy. The inclusion of redundant features not only increases computational complexity, but also degrades generalization capability. Hence, it is important to quantify and minimize redundancy among features during the selection process. Hence, this work computes pairwise redundancy among features using MI, which measures shared information between two variables. For any two features f_j and f_k, redundancy is defined as Eq. (2).

$$\mathcal{S}(f_j, f_k) = \mathcal{I}(f_j; f_k) \tag{2}$$

In Eq. (2), $\mathcal{I}(f_j; f_k)$ represents MI between features f_j and f_k. A higher value of $\mathcal{I}(f_j; f_k)$ indicates that two features are highly dependent and contain overlapping information. For evaluating overall redundancy linked with a specific feature f_j, the average redundancy is computed considering MI with all other features using Eq. (3).

$$Red(f_j) = \frac{1}{d-1} \sum_{k \neq} \mathcal{I}(f_j; f_k) \tag{3}$$

In Eq. (3), the term d represents the total number of features, and the summation is performed over all features except f_j itself. This ensures that each feature is evaluated in the context of entire feature space. A high value of $Red(f_j)$ shows that the feature shares significant information with other features and may be redundant. Conversely, a low redundancy value implies that the feature provides unique information. By incorporating these redundancy measures, the AWFOL-IDS avoids selecting features that do not contribute additional knowledge, thereby improving efficiency. After estimating both relevance using Eq. (1) and redundancy using Eq. (2) and Eq. (3), this work assigns dynamic and adaptive weights for each feature. This weight provides the overall importance of a feature by balancing its contribution to the target variable and its redundancy with other features. The weight assigned to feature f_j is defined using Eq. (4).

$$\mathcal{W}_j = \alpha \bullet \mathcal{R}(f_j) - \beta \bullet Red(f_j) \tag{4}$$

In Eq. (4), $\mathcal{R}(f_j)$ represents the relevance of feature f_j with respect to class label, $Red(f_j)$ represents the relevance of feature f_j with respect to the class label, $Red(f_j)$ represents the redundancy of feature f_j, α is the relevance coefficient, and β is the redundancy penalty coefficient. This ensures that features having high relevance and low redundancy receive higher weights, while redundant or less informative features are penalized. Unlike traditional approaches, which use fixed parameters, the AWIE-FS introduces adaptive parameter tuning for dynamically adjusting the influence of relevance and redundancy based on dataset characteristics using Eq. (5) and Eq. (6).

$$\alpha = \frac{\sigma_y}{\sigma_f + \in} \tag{5}$$

$$\beta = 1 - \alpha \tag{6}$$

In Eq. (5), σ_y represents the variance of class labels, representing class diversity, σ_f represents the average variance across all features and ? represents a small constant added to avoid division by zero. This adaptive formulation has important implications, i.e., when class variance σ_y is high (i.e., more diversity in class labels), AWIE-FS emphasizes feature relevance, increasing α and when feature variance σ_f is high, AWIE-FS increases the penalty on redundancy by adjusting β accordingly. Thus, the weighting approach becomes data-driven, allowing AWFOL-IDS to automatically adapt to different datasets without manual tuning. Once feature weights are computed, the next challenge is determining which features should be selected. Instead of using a fixed threshold or selecting a predefined number of features, the AWIE-FS introduces a dynamic thresholding approach that adapts to the distribution of feature weights. The threshold T is defined as Eq. (7).

$$\mathcal{T} = \mu_{\mathcal{W}} + \lambda \bullet \sigma_{\mathcal{W}} \tag{7}$$

In Eq. (7), $\mu_{\mathcal{W}}$ is the mean of all feature weights, $\sigma_{\mathcal{W}}$ is the standard deviation of feature weights, and λ is a tunable parameter controlling selection. This method ensures that only features having significantly higher importance than average are selected. The inclusion of the standard deviation term allows the threshold to adjust based on the spread of weights, i.e., if weights are tightly clustered, the threshold remains close to mean, and if weights are widely distributed, the threshold becomes more selective. The final selected feature subset is defined as Eq. (8).

$$\mathcal{F}^* = \{f_j | \mathcal{W}_j > \mathcal{T}\} \tag{8}$$

Equation (8) results in a data-driven feature subset that automatically adapts to the underlying distribution of feature importance. The adaptive feature selection process plays a crucial role in improving the efficiency and performance of AWFOL-IDS by ensuring that only the most important features are used for prediction and classification stages, as discussed in the next section.

3.4 Classifier

To achieve better performance while maintaining computational efficiency, this work presents WARF-FIO. The WARF-FIO classifier extends conventional RF by incorporating three key improvements. The first is adaptive feature weighting derived from the feature selection stage, the second is class imbalance-aware learning using sample reweighting, and finally dynamic tree-level optimization for improving ensemble performance. These modifications enable the classifier to achieve better prediction/classification performance while remaining computationally efficient for large-scale IoT environments. Consider the optimal feature subset obtained from the feature selection stage as $\mathcal{F}^* = \{f_1, f_2, \ldots, f_m\}$, where $m \ll d$ which ensures dimensionality reduction and computational efficiency. An RF model usually consists of an ensemble of T decision trees, where each tree independently learns a mapping from input features to output classes. The overall prediction of the ensemble is computed as Eq. (9).

$$F(x) = \frac{1}{T} \sum_{t=1}^{T} h_t(x) \tag{9}$$

In Eq. (9), x is the input feature vector, $h_t(x)$ represents the prediction of tth decision tree and T represents the total number of trees in the forest. In traditional RF, each tree contributes equally to the final prediction. However, this assumption may not be optimal when trees exhibit varying levels of performance, particularly in imbalanced datasets. In standard RF, node splitting is performed using impurity measures like the Gini index, as in Eq. (10).

$$\mathcal{G} = 1 - \sum_{c=1}^{K} p_c^2 \tag{10}$$

In Eq. (10), p_c represents the probability of class c at a given node. The WARF-FIO modifies this criterion by incorporating adaptive feature weights $\mathcal{W}_j$ obtained from the feature selection stage. The weighted Gini index is defined as Eq. (11).

$$\mathcal{G}_w = \sum_{j \in \mathcal{F}^*} \mathcal{W}_j \left(1 - \sum_{c=1}^{K} p_{c|j}^2\right) \tag{11}$$

In Eq. (11), $\mathcal{W}_j$ represents the importance of feature f_j, $p^2_{c|j}$ represents the conditional probability of class c given feature j. This method ensures that features having higher discriminative power exert more influence during the split selection process. As a result, the constructed trees are more informative and better aligned with the underlying data distribution. Further, as seen in Sect. 3.1, the IoT and IoMT datasets are highly imbalanced, with attack samples significantly outnumbering benign samples. For addressing the following issue, a class-dependent sample weight approach is introduced. Each sample training x_i is assigned a weight using Eq. (12).

$$\omega_i = \frac{\mathcal{N}}{K \bullet \mathcal{N}_{y_i}} \tag{12}$$

In Eq. (12), $\mathcal{N}$ represents the total number of samples, K is the number of classes, and $\mathcal{N}_{y_i}$ represents the number of samples belonging to a class y_i. This formulation assigns higher weights to samples from minority classes, thereby compensating for class imbalance. The class probabilities at each node are then computed using weighted frequencies using Eq. (13).

$$p_c = \frac{\sum_{i \in c} \omega_i}{\sum_{i=1}^{\mathcal{N}} \omega_i} \tag{13}$$

Equation (13) ensures that minority classes contribute proportionally more during the tree-building process, improving detection sensitivity for rare attack types. Further, traditional RF employs uniform bootstrap sampling, where each sample has an equal probability of being selected. However, this approach is not suitable for imbalanced datasets. Hence, for overcoming the following, the AWFOL-IDS uses weighted bootstrap sampling, where the selection probability of each sample is proportional to its weight, i.e., by using Eq. (14).

$$\mathcal{P}(x_i) = \frac{\omega_i}{\sum_{k=1}^{\mathcal{N}} \omega_k} \tag{14}$$

This adaptive sampling strategy increases the likelihood of selecting minority class samples during training, thereby improving class representation in each decision tree. Consequently, the ensemble becomes more balanced and capable of detecting both majority and minority class patterns effectively. Unlike conventional RF, where all trees contribute equally, the WARF-FIO assigns performance-based weight to each tree. The weight of tth is defined using Eq. (15).

$$\alpha_t = \frac{1}{\in_t + \delta} \tag{15}$$

In Eq. (15), $?_t$ denotes the classification error of the tree t and δ is a small constant to avoid division by zero. Trees with lower error receive higher weights, thereby contributing more significantly to the final prediction and classification. The ensemble output is computed using Eq. (16).

$$F(x) = \frac{\sum_{t=1}^{T} \alpha_t h_t(x)}{\sum_{t=1}^{T} \alpha_t} \tag{16}$$

This weighted aggregation improves AWFOL-IDS performance by emphasizing high-performing trees. Further, to prevent overfitting and improve generalization, this work incorporated a regularization approach into the learning process. Hence, the objective function is defined as Eq. (17).

$$\mathcal{J} = \sum_{t=1}^{T} \alpha_t \mathcal{L}_t + \lambda \cdot \Omega \tag{17}$$

In Eq. (17), $\mathcal{L}_t$ represents the loss linked to the tree t, λ represents the regularization coefficient, and Ω represents the feature weight regularization term. The regularization term is defined as in Eq. (18).

$$\Omega = \sum_{j=1}^{m} \mathcal{W}_j^2 \tag{18}$$

The Eq. (18) term penalizes excessively large feature weights, ensuring that AWFOL-IDS does not over-rely on a small subset of features. As a result, the classifier maintains a balanced representation of the feature space while avoiding overfitting. The final prediction/classification is obtained by selecting the class with the maximum aggregated score, as defined in Eq. (19)

$$\hat{y} = \arg \max_{c \in \{1,\ldots K\}} F_c(x) \tag{19}$$

In Eq. (19), $F_c(x)$ represents the predicted score for the class c. Hence, this WARF-FIO classifier improves traditional RF by incorporating adaptive feature weighting, imbalance-aware sampling, and tree-level optimization. These improvements enable AWFOL-IDS to achieve high detection and classification accuracy while maintaining low computational complexity. The overall process of the AWFOL-IDS is presented in Algorithm 1, as discussed in the next section.

3.5 AWFOL-IDS

This section presents the complete process of AWFOL-IDS using Algorithm 1 given below. Algorithm 1 begins by loading the dataset (Step 2) and performing data preprocessing (Step 3-Step 5). Next, feature relevance is computed using MI (Step 6), followed by redundancy estimation using pairwise MI among features (Step 7). Based on these, adaptive weights are assigned to each feature utilizing relevance-redundancy trade-off parameters derived from the dataset (Step 8-Step 14). A dynamic threshold is then calculated for selecting the optimal feature subset (Step 15-Step 17). To address class imbalance, class-wise sample counts are computed, and weights are assigned inversely proportional to class frequency (Step 18-Step 19). These weights are used in weighted bootstrap sampling for generating balanced training subsets (Step 20). A set of decision trees is constructed using selected features, where each tree is trained using the weighted Gini index for optimal splitting (Step 21). Tree errors are computed, and adaptive weights are assigned to each tree. During prediction, outputs from all trees are aggregated using weighted voting. Finally, AWFOL-IDS performance is evaluated using standard metrics (Step 24).

Input	Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, number of trees T, regularization coefficient λ, constant ϵ_t, δ
Output	Trained WARF-FIO, predicted class labels for prediction/classification $\hat{y}$
Step 1	Begin
Step 2	Load dataset $\mathcal{D}$
Step 3	For each feature f_j in $\mathcal{D}$ do
	If f_j contains missing values, then
	If f_j is numerical then
	Replace missing values using the median (f_j)
	Else
	Replace missing values using the mode (f_j)
	End if
	End if
	End for
Step 4	For each categorical feature f_j do
	If cardinality f_j is low then
	Apply one-hot encoding
	Else
	Apply frequency encoding
	End if
	End for
Step 5	Remove constant and highly correlated features
Step 6	For each feature f_j do
	Compute Eq. (1).
	End for
Step 7	For each feature f_j do
	$Red(f_j) = 0$
	For each feature f_k where $k \neq j$ do
	$Red(f_j) = Red(f_j) + \mathcal{I}(f_j; f_k)$
	End for
	$Red(f_j) = \frac{Red(f_j)}{(d-1)}$
	End for
Step 8	Compute $\sigma_y = varaince(y)$
Step 9	Compute $\sigma_f = average\ of\ all\ features$
Step 10	Compute Eq. (5)
Step 11	Compute Eq. (6)
Step 12	For each feature f_j do
	Compute Eq. (4)
	End for
Step 13	Compute $\mu_W = mean(\mathcal{W}_j)$ (mean of Eq. (4))
Step 14	Compute $\sigma_W = standard\ deviation\ (\mathcal{W}_j)$ (standard deviation of Eq. (4))
Step 15	Compute Eq. (7)
Step 16	Initialize feature set $\mathcal{F}^* = \{\}$
Step 17	For each feature f_j do
	If $\mathcal{W}_j$(Eq. 4)> $\mathcal{T}$(Eq. 7) then
	Add f_j to $\mathcal{F}^*$ (Eq. 8)
	End If
	End for
Step 18	For each class c do
	$\mathcal{N}_c = number\ of\ samples\ in\ class\ c$
	End for
Step 19	For each sample x_i do
	Compute Eq. (12)
	End for
Step 20	For $t = 1$ to T do
	Create bootstrap sample D_t using likelihood Eq. (14)
	Train decision tree h_t on the decision tree using features $\mathcal{F}^*$ (Eq. (8))
	While true h_t is not fully grown, do
	For each candidate split feature $f_j \in \mathcal{F}^*$ do
	Compute weighted Gini $\mathcal{G}_w$ (Eq. (11))
	End for
	Select feature with minimum $\mathcal{G}_w$ (Eq. (11))
	Split node
	End While
	Compute error of tree h_t
	Compute tree weight α_t
	End for
Step 21	For each test sample x do
	$\alpha_t h_t(x), \alpha_t = 0 = 0$
	For $t = 1$ to T do
	$prediction = ht(x)$
	Compute $\sum_{t=1}^{T} \alpha_t\, h_t(x)$
	Compute $\sum_{t=1}^{T} \alpha_t$
	End for
	Compute Eq. (16)
	Compute Eq. (19)
	End for
Step 22	Return trained WARF-FIO
Step 23	Return predicted labels $\hat{y}$
Step 24	Evaluate performance using accuracy, precision, recall, and F-score.
Step 25	Stop

Algorithm 1. AWFOL-IDS

Using the following approach, the AWFOL-IDS is evaluated, as discussed in detail in the next section.

4 Results and discussion

The AWFOL-IDS was developed on a system with an Intel i3 processor, a 500 GB SSD, and 8 GB RAM. For the evaluation of the study, the datasets discussed in Sect. 3.1, i.e., CICIoT2023 [11, 25] and CICIoMT2024 [12, 26]. In this work, the work was evaluated using an 80:20 train-test ratio, as most of the existing works have used a similar ratio for training and testing their approaches [11, 12, 17, 20] and [21]. For a comparative study, similar studies have been considered. This section first discusses the performance metrics used for evaluating AWFOL-IDS. Further, the AWFOL-IDS was evaluated using CICIoT2023 and CICIoMT2024 for both prediction (binary-class) and classification (multi-class (8 categories and 34 classes for CICIoT2023) and (6 categories and 19 classes)). Followed by which this section provides a cross dataset evaluation considering similar classes from the CICIoT2023 and CICIoMT2024 datasets. Furthermore, a comparative study is conducted, where similar training-testing ratio studies were considered for comparison. Finally, this study provides an ablation study, discussing how each process, i.e., preprocessing, feature selection, and classifier components, shows different performance, i.e., when considered and when not considered. The simulation setup and hyperparameter selection presented in Table 8 were performed using 5-fold Cross-Validation (CV) on the training partition, so that the test-set results reported in Sect. 4 were not used at any stage of feature selection or hyperparameter tuning.

4.1 Performance metrics

For the evaluation of AWFOL-IDS, this work considered the standard performance metrics, i.e., accuracy, precision, recall, F-score, and AUC-ROC, as most of the existing works discussed in the literature survey also have used similar metrics. The following metrics are evaluated using Eqs. (20)-Eq. (24), where TP represents true-positive, FP represents false-positive, TN represents true-negative, FN represents false-negative, TPR represents the true-positive rate and FPR represents the false-positive rate.

Table 8 Simulation setup and hyperparameter selection of the proposed AWFOL-IDS model

Category	Parameter	Value / Range
Hardware Environment	Processor	Intel i3
	RAM	8 GB
	Storage	500 GB SSD
Datasets	CICIoT2023	46,686,579 samples (34 attack classes, 8 categories)
	CICIoMT2024	8,775,013 samples (19 attack classes, 6 categories)
Data Partitioning	Train-Test Split	80:20 (stratified)
	Validation Strategy	Stratified 5-Fold CV (on training partition)
AWIE-FS (Feature Selection)	Relevance coefficient (α)	Adaptive range $[0.55, 0.82]$
	Redundancy penalty coefficient (β)	Adaptive range $[0.18, 0.45]$
	Threshold tunable parameter (k, Eq. 7)	$1.0\,(\mu + k \cdot \sigma)$
WARF-FIO (Classifier)	Number of trees (T)	100
	Maximum tree depth	20
	Minimum samples per split	5
	Minimum samples per leaf	2
	Regularization coefficient (λ)	Tuned using 5-Fold CV Grid Search: $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1.0\}$ (Stratified 5-Fold CV) Selected: 0.01
	Smoothing constants (ϵ, δ)	10^{-5}
Evaluation Metrics	Accuracy, Precision, Recall, F-Score, AUC-ROC	-

$$Accuracy = \left(\frac{TP + TN}{TP + TN + FP + FN}\right) \tag{20}$$

$$Precision = \left(\frac{TP}{TP + FP}\right) \tag{21}$$

$$Recall = \left(\frac{TP}{TP + FN}\right) \tag{22}$$

$$F - Score = \left(2 \bullet \frac{Precision \bullet Recall}{Precision + Recall}\right) \tag{23}$$

$$AUC = \left(\int TPRd(FPR)\right) \tag{24}$$

4.2 Performance evaluation of CICIoT2023

This section presents the performance evaluation of AWFOL-IDS on the CICIoT2023 dataset for both binary-class and multi-class. As presented in Fig. 2, the AWFOL-IDS achieves better results for 2-class (0.9989 accuracy, 0.9975 precision, 0.9968 recall and 0.9971 f-score), 8-class (0.9968 accuracy, 0.9923 precision, 0.9925 recall and 0.9918 f-score) and 34-class (0.9939 accuracy, 0.9986 precision, 0.9952 recall and 0.9994 f-score). The confusion matrix and AUC-ROC curve for 2-class, 8-class, and 34-class are given from Figs. 3, 4, 5, 6, 7 and 8. The performance achieved by AWFOL-IDS is due to the adaptive feature weighting approach defined in Eq. (4), which helps in balancing relevance and redundancy, and dynamic thresholding in Eq. (7) for selecting optimal features. Also, the weighted Gini index in Eq. (11) and class imbalance handling using sample weighting in Eq. (12) improve classification outcomes for all different classes. Moreover, the final prediction using weighted ensemble aggregation in Eq. (16) improves reliability, contributing to the accuracy and F-score improvements reported in Fig. 2 and Tables 9, 10 relative to the RF and DNN baselines. Also, the per-class results achieved by AWFOL-IDS on CICIoT2023 2-class, 8-class, and 34-class are presented in Tables 9 and 10, and Table 11.

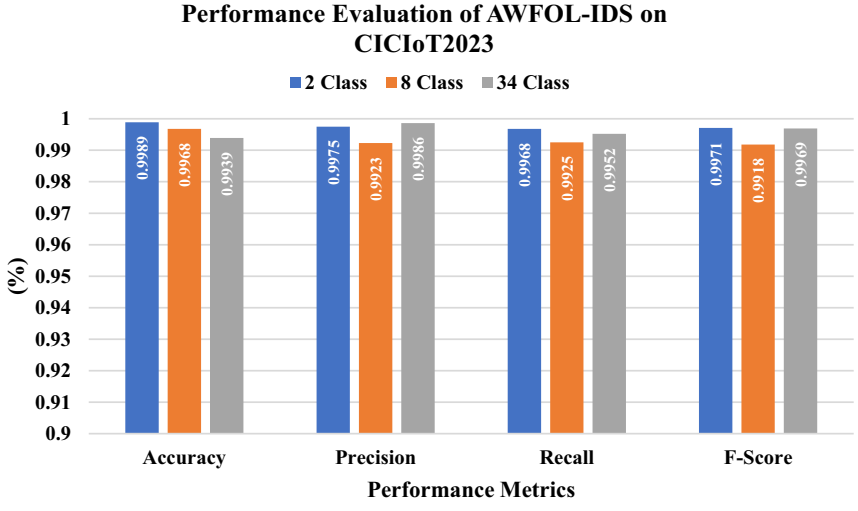


Fig. 2 AWFOL-IDS Evaluation on CICIoT2023

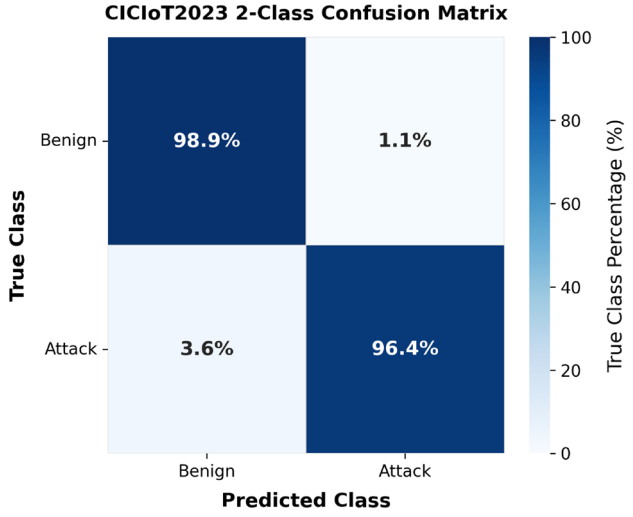


Fig. 3 Confusion Matrix of AWFOL-IDS Evaluation on CICIoT2023 for 2 classes

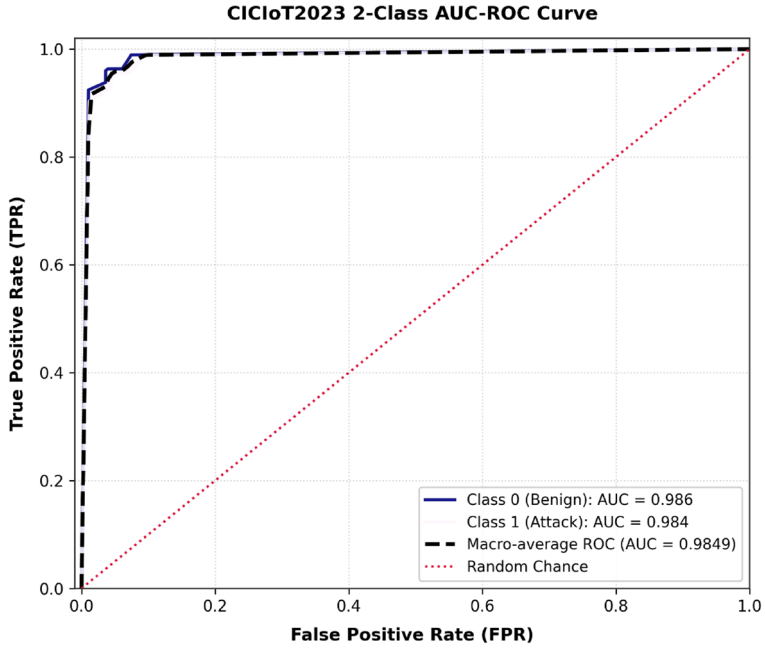


Fig. 4 AUC-ROC of AWFOL-IDS Evaluation on CICIoT2023 for 2 classes

4.3 Performance evaluation of CICIoMT2024

This section evaluates the performance of AWFOL-IDS on CICIoMT2024 datasets. The results achieved for different classes are presented in Fig. 9, where AWFOL-IDS shows better outcomes for binary and multi-class classification tasks. For 2 classes, AWFOL-IDS achieved 0.9998 accuracy, 0.9932 precision, 0.9991 recall, and 0.9961 F-score. For 6 classes, the AWFOL-IDS achieved 0.9992 accuracy, 0.9989 precision, 0.999 recall, and 0.9989 F-score. For 19 classes, the AWFOL-IDS achieved 0.9856 accuracy, 0.9878 precision, 0.9972 recall, and 0.9925 F-score.The confusion matrix and AUC-ROC curve for 2-class,6-class and 19-class are given from figure 10 to 15. This performance is achieved because of the adaptive feature weighting approach defined in Eq. (4), which selects informative features while reducing redundancy, along with dynamic thresholding in

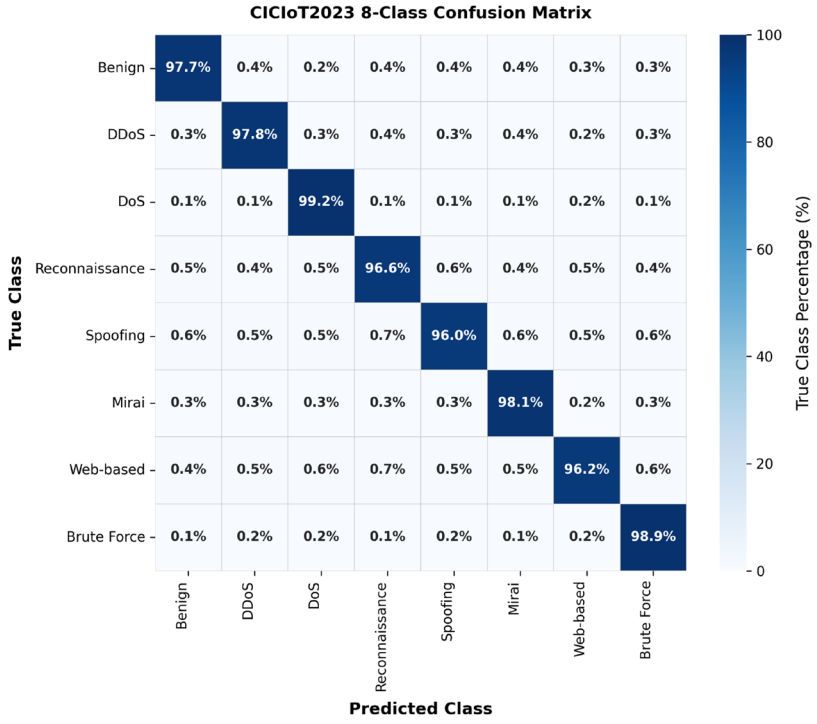


Fig. 5 Confusion Matrix of AWFOL-IDS Evaluation on CICIoT2023 for 8 classes

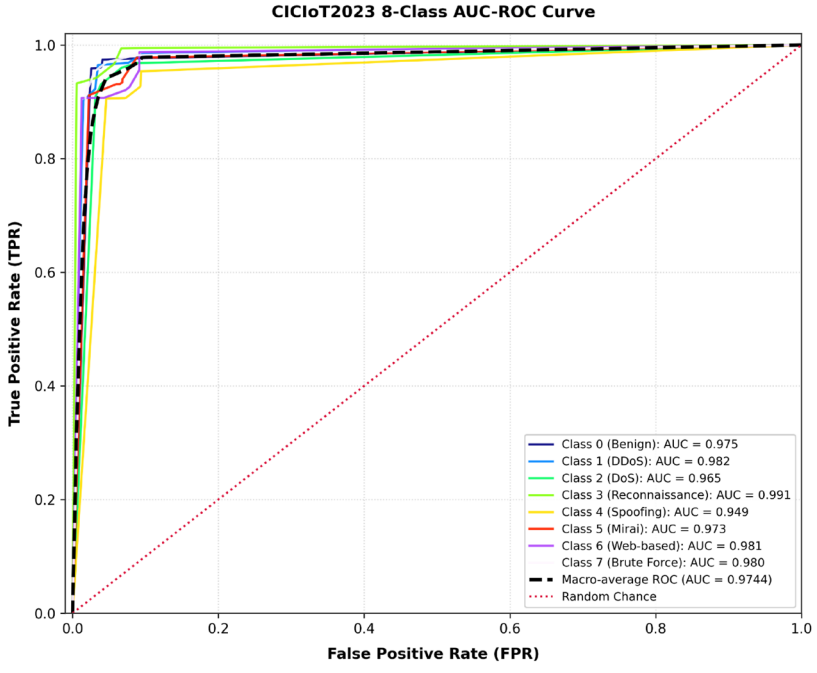


Fig. 6 AUC-ROC of AWFOL-IDS Evaluation on CICIoT2023 for 8 classes

Eq. (7). Additionally, Eq. (11) improves decision tree splitting, while class imbalance was effectively handled using sample weighting in Eq. (12). The final prediction aggregation was done using Eq. (16), which improved overall performance, enabling AWFOL-IDS to achieve high detection accuracy across IoMT environments. The per-class results

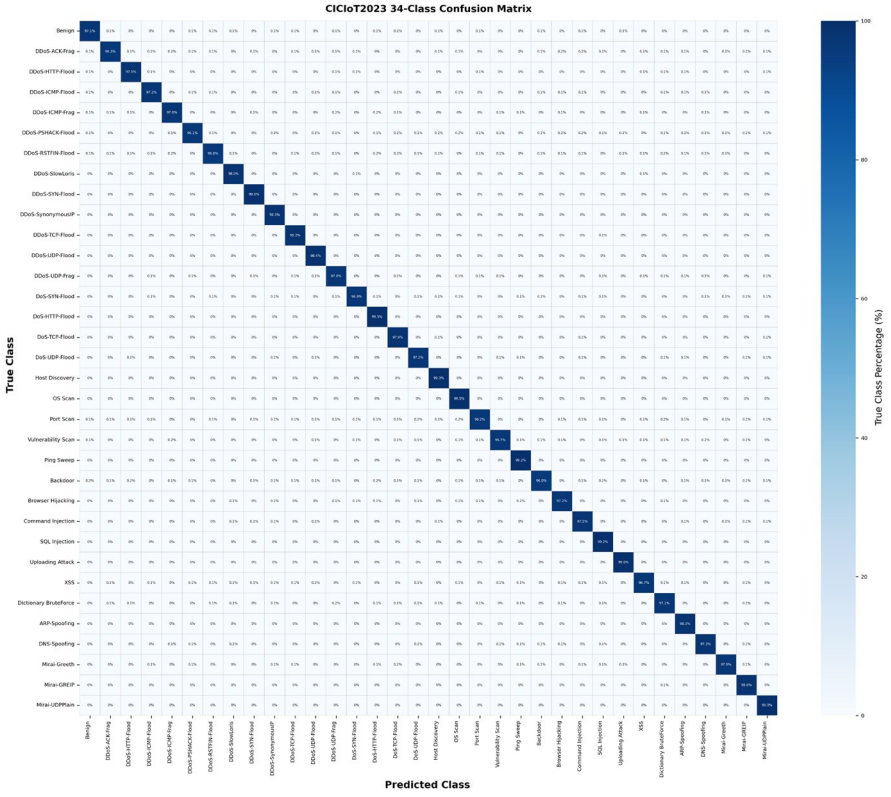


Fig. 7 Confusion Matrix of AWFOL-IDS Evaluation on CICIoT2023 for 34 classes

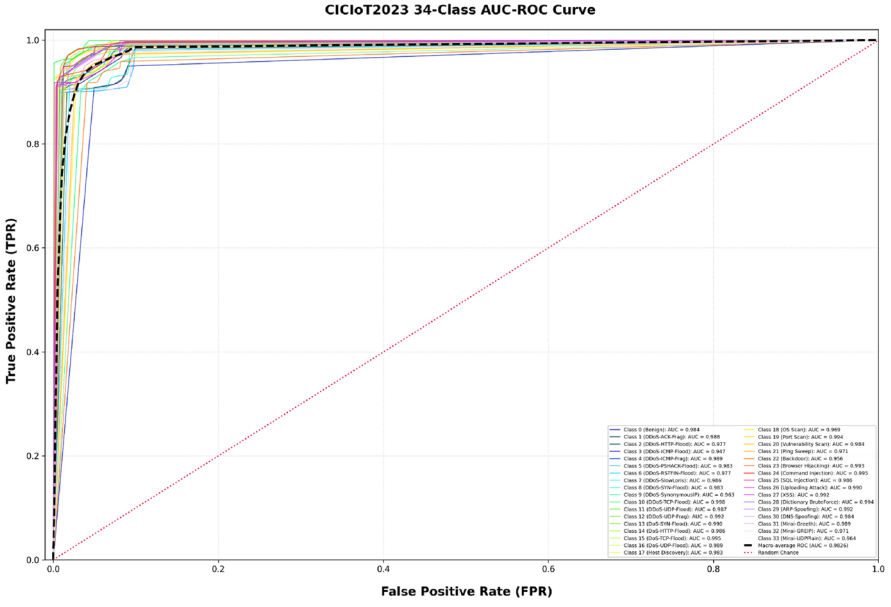


Fig. 8 AUC-ROC of AWFOL-IDS Evaluation on CICIoT2023 for 34 classes

Table 9 Per-class evaluation of AWFOL-IDS on CICIoT2023 for 2 classes

Class	Precision	Recall	F-score	Support
Benign	0.9982	0.9961	0.9971	219,639
Attack	0.9968	0.9975	0.9971	9,117,677
Macro Average	0.9975	0.9968	0.9971	9,337,316
Weighted Average	0.9968	0.9975	0.9971	9,337,316

Table 10 Per-class evaluation of AWFOL-IDS on CICIoT2023 for 8 classes

Class	Precision	Recall	F-score	Support
Benign	0.9982	0.9961	0.9971	219,639
DDoS	0.9928	0.9959	0.9943	6,796,912
DoS	0.993	0.9928	0.9929	1,618,148
Reconnaissance	0.9953	0.9912	0.9932	70,913
Web-based	0.9856	0.9818	0.9837	4966
Brute Force	0.9987	0.9912	0.9949	2613
Spoofing	0.9892	0.9932	0.9912	97,301
Mirai	0.9856	0.9978	0.9917	526,825
Macro Average	0.9923	0.9925	0.9918	9,337,317
Weighted Average	0.9928	0.9951	0.9939	9,337,317

achieved by AWFOL-IDS on CICIoMT2024 2-class, 6-class, and 19-class are presented in Tables 12 and 13, and Table 14; Figs. 9, 10, 11, 12, 13, 14 and 15.

4.4 Cross dataset evaluation

For evaluation of the generalization capability of AWFOL-IDS, this work conducted cross-dataset experiments using CICIoT2023 and CICIoMT2024 datasets. Since both datasets contain different attack distributions, only common attack categories and classes were considered to ensure fair evaluation. The similar categories considered for this study included DDoS, DoS, Reconnaissance, and Spoofing, as these categories are present in both datasets, as presented in Table 7. The evaluation was performed using an 80:20 training-testing ratio, where the AWFOL-IDS was trained on one dataset and tested on the other. Because CICIoT2023 and CICIoMT2024 differ substantially in their domain-specific attack types, for example, CICIoMT2024 includes MQTT-based attacks with no counterpart in CICIoT2023, while CICIoT2023 includes web-based, brute-force, and Mirai-based attacks absent from CICIoMT2024, the cross-dataset generalization claim in this work is scoped specifically to the attack categories shared between the two datasets (DDoS, DoS, Reconnaissance, and Spoofing) rather than to the full attack surface of either environment. Evaluating generalization to domain-specific, non-overlapping attack types is left as future work.

The results show that AWFOL-IDS achieves better performance when trained and tested on different datasets, as presented in Tables 15 and 16. When trained on CICIoT2023 and tested on CICIoMT2024, the AWFOL-IDS achieved 0.9924 accuracy and when trained on CICIoMT2024 and tested on CICIoT2023, it achieved 0.9891 accuracy. The strong cross-dataset performance is because of the adaptive feature selection approach defined in Eq. (4), ensuring that only highly relevant and non-redundant features were selected, showing that AWFOL-IDS is less dependent on dataset-specific characteristics. Additionally, the dynamic thresholding in Eq. (7) allowed the feature selection process to adapt to variations in feature distributions across datasets. The classifier further improves generalization using the weighted Gini index in Eq. (11),

Table 11 Per-class evaluation of AWFOL-IDS on CICIoT2023 for 34 classes

Class	Precision	Recall	F-score	Support
Benign	0.9982	0.9961	0.9971	219,639
DDoS-ACK-Fragmentation	0.9995	0.9992	0.9993	57,021
DDoS-HTTP-Flood	0.9988	0.9963	0.9975	5758
DDoS-ICMP-Flood	0.9998	0.9998	0.9998	1,440,101
DDoS-ICMP-Fragmentation	0.9993	0.9994	0.9993	90,498
DDoS-PSHACK-Flood	0.9995	0.9991	0.9993	818,951
DDoS-RSTFIN-Flood	0.9994	0.9992	0.9993	809,057
DDoS-Slow Loris	0.9987	0.9972	0.9979	4685
DDoS-SYN-Flood	0.9996	0.9995	0.9995	811,838
DDoS-Synonymous IP-Flood	0.9992	0.9991	0.9991	719,628
DDoS-TCP-Flood	0.9996	0.9993	0.9994	899,533
DDoS-UDP-Flood	0.9997	0.9998	0.9997	1,082,457
DDoS-UDP-Fragmentation	0.9993	0.9994	0.9993	57,385
DoS-SYN-Flood	0.9995	0.9993	0.9994	405,767
DoS-HTTP-Flood	0.9986	0.9962	0.9974	14,373
DoS-TCP-Flood	0.9994	0.9991	0.9992	534,289
DoS-UDP-Flood	0.9998	0.9997	0.9997	663,719
Host Discovery	0.9982	0.9928	0.9955	26,876
OS Scan	0.9984	0.9931	0.9957	19,652
Port Scan	0.9981	0.9918	0.9949	16,457
Vulnerability Scan	0.9979	0.9902	0.994	7476
Ping Sweep	0.998	0.9882	0.9931	452
Backdoor-Malware	0.9976	0.988	0.9928	644
Browser Hijacking	0.9972	0.9851	0.9911	1172
Command Injection	0.9978	0.9862	0.992	1082
SQL Injection	0.9971	0.9848	0.9909	1049
Uploading Attack	0.9975	0.9858	0.9916	250
XSS	0.9973	0.9852	0.9912	769
Dictionary Brute Force	0.9987	0.9912	0.9949	2613
ARP-Spoofing	0.9985	0.9952	0.9968	61,519
DNS-Spoofing	0.998	0.9912	0.9946	35,782
Mirai-Greeth Flood	0.9992	0.9987	0.9989	198,373
Mirai-GREIP Flood	0.9991	0.9988	0.9989	150,336
Mirai-UPD Plain	0.9993	0.9989	0.9991	178,115
Macro Average	0.9986	0.9952	0.9969	9,337,316
Weighted Average	0.9996	0.9996	0.9996	9,337,316

which prioritizes important features during tree construction, and the class imbalance handling approach in Eq. (12), ensuring fair representation of all classes. Finally, the weighted ensemble aggregation in Eq. (16) improves prediction outcomes across different data distributions. These improvements made in the proposed AWFOL-IDS help to effectively learn generalized attack patterns, which results in high accuracy in cross-dataset evaluation.

4.5 Comparative study

The comparative study presented in Tables 17 and 18 highlights limitations of existing approaches and shows the performance of AWFOL-IDS across both CICIoT2023 and CICIoMT2024 datasets. From Table 17, it is observed that traditional models like LR, MLP, and AB show significantly lower performance, especially in multi-class scenarios (8-class and 34-class), due to their inability to effectively capture complex attack patterns and handle high-dimensional data. Even though RF and DNN achieve relatively higher

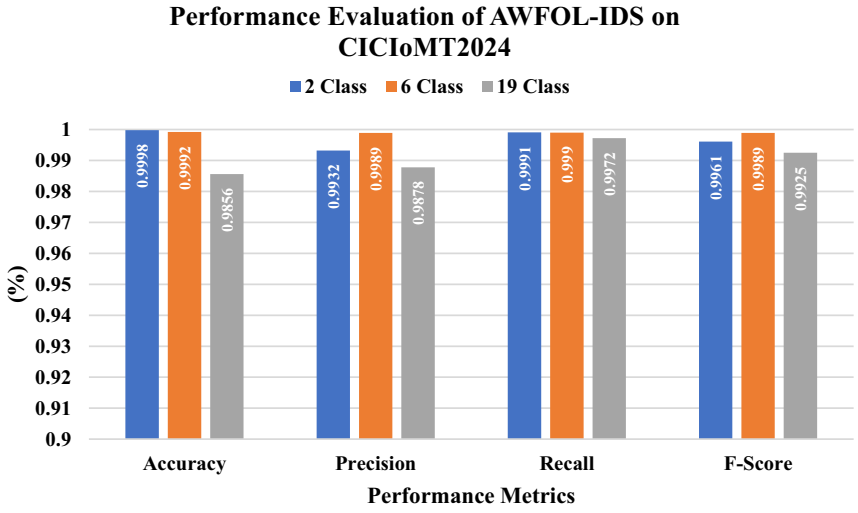


Fig. 9 AWFOL-IDS evaluation on CICIoMT2024

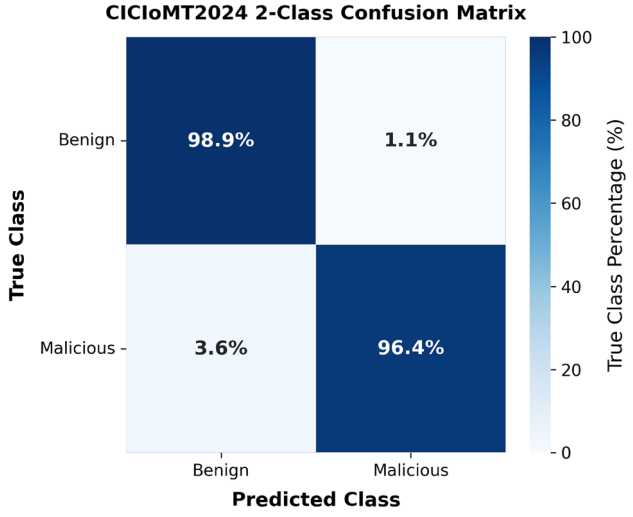


Fig. 10 Confusion Matrix of AWFOL-IDS Evaluation on CICIoMT2024 for 2 classes

accuracy, their precision, recall, and F-score values indicate inconsistency in correctly identifying minority classes, primarily due to class imbalance and redundant feature inclusion. Similarly, feature selection-based approaches in [21], such as PCA and χ^2, either reduce performance or provide only marginal improvements, as they rely on static selection strategies without adapting to data characteristics.

A similar trend is observed in Table 18 for the CICIoMT2024 dataset, where conventional ML and DL models struggle in multi-class classification. While some advanced models like Transformer and GCN show improved results, they introduce high computational complexity, making them less suitable for real-time IoMT environments. Additionally, many of these approaches fail to explicitly address class imbalance, leading to reduced recall and F-score, particularly for minority attack classes. In contrast, AWFOL-IDS consistently achieves superior performance across all class levels. For CICIoT2023, it achieves 0.9989, 0.9968, and 0.9939 accuracy for 2-class, 8-class, and 34-class, respectively, with corresponding high precision, recall, and F-score values. Similarly, for

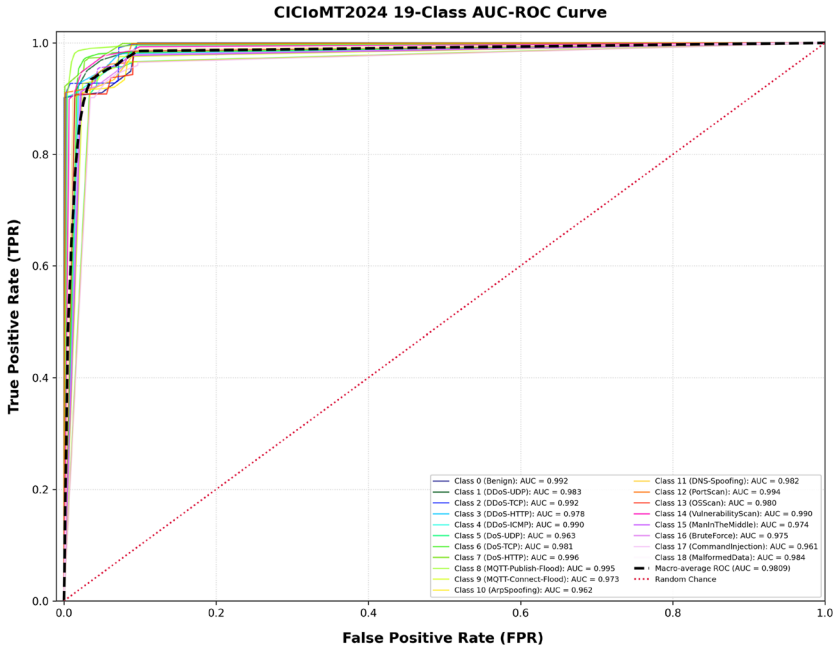


Fig. 15 AUC-ROC of AWFOL-IDS Evaluation on CICIoMT2024 for 19 classes

Table 12 Per-class evaluation of AWFOL-IDS on CICIoMT2024 for 2 classes

Class	Precision	Recall	F-score	Support
Benign	0.9882	0.9989	0.9935	46,068
Attack	0.9982	0.9993	0.9987	1,708,935
Macro Average	0.9932	0.9991	0.9961	1,755,003
Weighted Average	0.9977	0.9993	0.9985	1,755,003

Table 13 Per-class evaluation of AWFOL-IDS on CICIoMT2024 for 6 classes

Class	Precision	Recall	F-score	Support
Benign	0.9882	0.9989	0.9935	46,068
DDoS	0.9998	0.9995	0.9996	1,169,325
DoS	0.9996	0.9991	0.9993	444,441
Reconnaissance	0.9982	0.9985	0.9983	26,280
Spoofing	0.9981	0.9991	0.9986	3558
MQTT	0.9988	0.998	0.9984	65,331
Macro Average	0.9989	0.999	0.9989	1,755,003
Weighted Average	0.9989	0.9992	0.9991	1,755,003

CICIoMT2024, it achieves 0.9998, 0.9992, and 0.9856 accuracy for 2-class, 6-class, and 19-class scenarios.

This improvement is because of the adaptive feature weighting mechanism in Eq. (4), which ensures selection of highly relevant and non-redundant features, along with dynamic thresholding in Eq. (7) that adapts to feature distribution. Furthermore, the weighted Gini index in Eq. (11) enhances decision tree learning, while class imbalance handling using Eq. (12) improves detection of minority classes. The weighted ensemble aggregation in Eq. (16) further strengthens prediction reliability. These combined mechanisms enable AWFOL-IDS to outperform existing methods while maintaining computational efficiency. Moreover, the studies summarized in Tables 17 and 18 do not

Table 14 Per-class evaluation of AWFOL-IDS on CICIoMT2024 for 19 classes

Class	Precision	Recall	F-score	Support
Benign	0.9882	0.9989	0.9935	46,068
DDoS-SYN Flood	0.9998	0.9995	0.9996	194,872
DDoS-TCP Flood	0.9997	0.9994	0.9995	197,413
DDoS-ICMP Flood	0.9998	0.9996	0.9997	377,435
DDoS-UDP Flood	0.9998	0.9996	0.9997	399,605
DoS-SYN Flood	0.9996	0.9991	0.9993	108,100
DoS-TCP Flood	0.9995	0.999	0.9992	92,496
DoS-ICMP Flood	0.9997	0.9992	0.9994	102,945
DoS-UDP Flood	0.9996	0.9991	0.9993	140,901
Ping Sweep	0.9852	0.9958	0.9905	185
Vulnerability Scan	0.9848	0.9952	0.99	641
OS Scan	0.9842	0.9948	0.9895	4133
Port Scan	0.985	0.9955	0.9902	21,321
ARP Spoofing	0.9981	0.9991	0.9986	3558
Malformed Data	0.9825	0.9942	0.9883	1375
DoS Connect Flood	0.982	0.9938	0.9879	3181
DDoS Connect Flood	0.9822	0.994	0.9881	42,990
DoS Publish Flood	0.983	0.9945	0.9887	10,576
DDoS Publish Flood	0.9832	0.9947	0.9889	7208
Macro Average	0.9878	0.9972	0.9925	1,755,003
Weighted Average	0.9985	0.9991	0.9988	46,068

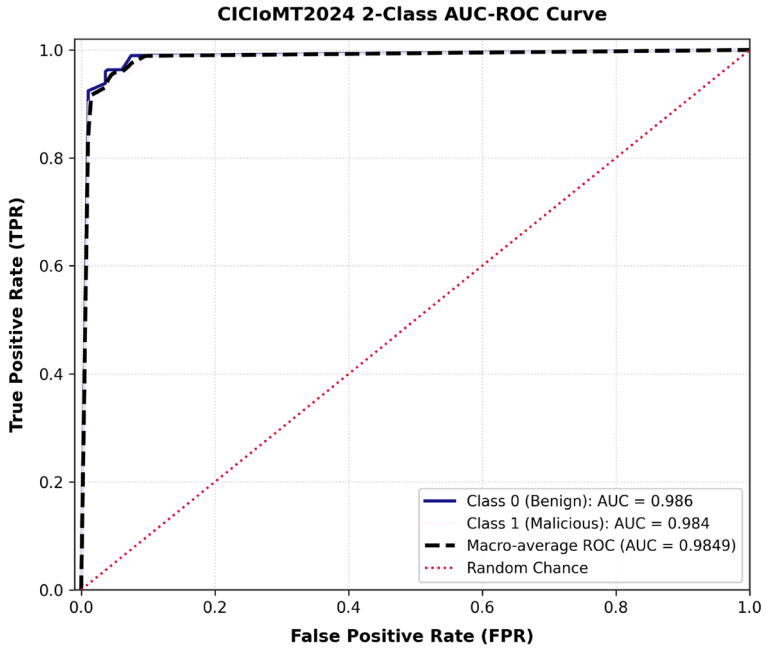


Fig. 11 AUC-ROC of AWFOL-IDS Evaluation on CICIoMT2024 for 2 classes

share an identical experimental protocol. Table 19 lists, for each compared study, the reported train-test ratio, the class grouping used for multi-class evaluation, and whether class-imbalance handling was applied, so that differences in reported accuracy can be interpreted in light of differing evaluation assumptions rather than treated as directly equivalent to the AWFOL-IDS protocol used in this work.

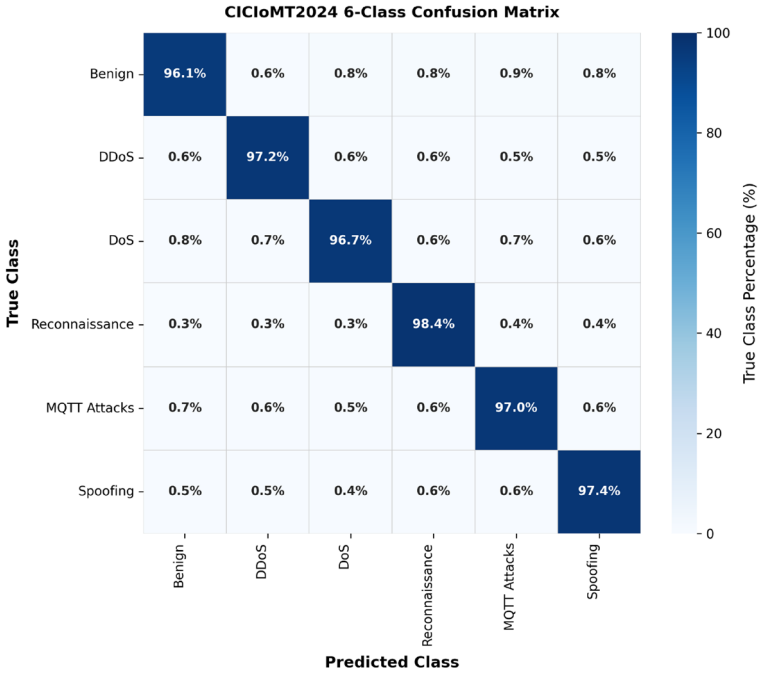


Fig. 12 Confusion Matrix of AWFOL-IDS Evaluation on CICIoMT2024 for 6 classes

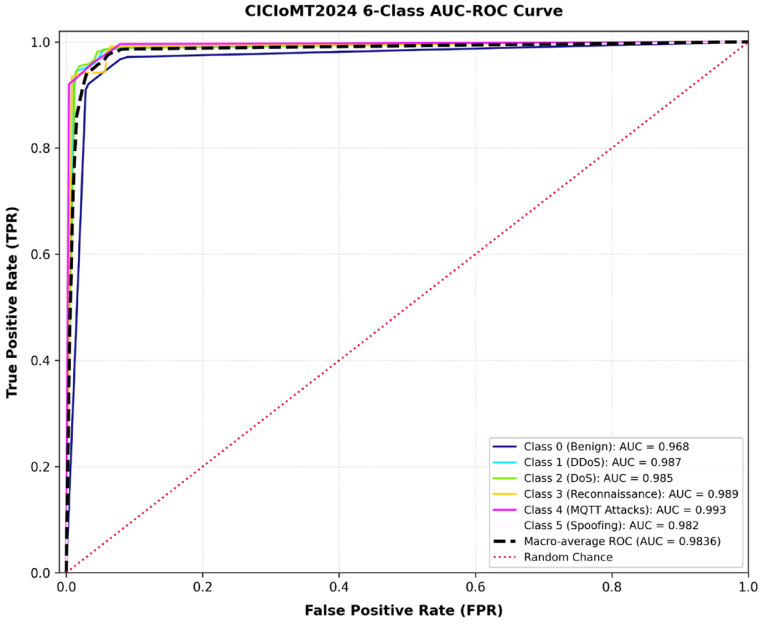


Fig. 13 AUC-ROC of AWFOL-IDS Evaluation on CICIoMT2024 for 6 classes

In the next section, the ablation study is presented, discussing how each different component affects AWFOL-IDS performance.

4.6 Computational complexity and resource profiling

To assess the practical deploy ability of AWFOL-IDS beyond classification accuracy, this subsection reports the computational complexity of AWFOL-IDS. Table 20 summarizes training time, average per-sample inference latency, peak memory consumption during

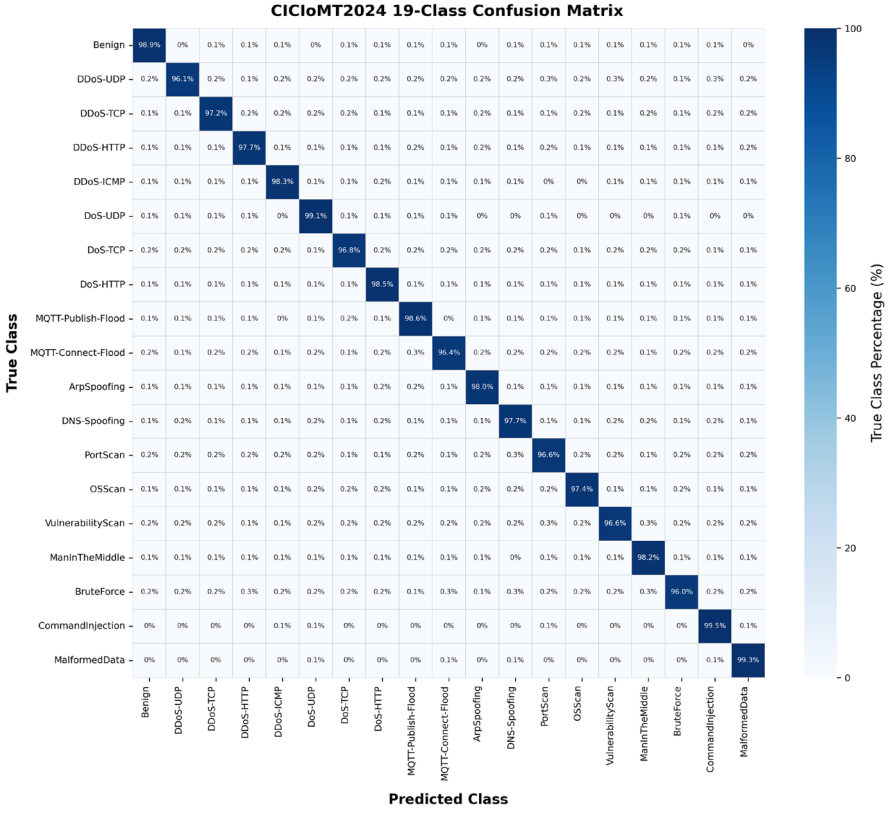


Fig. 14 Confusion Matrix of AWFOL-IDS Evaluation on CICIoMT2024 for 19 classes

Table 15 AWFOL-IDS Performance Evaluation of CICIoMT2024 when trained on CICIoT2023

Class type	Accuracy	Precision	Recall	F-Score
4 Class (DDoS, DoS, Reconnaissance and Spoofing)	0.9924	0.9912	0.9936	0.9924

Table 16 AWFOL-IDS Performance Evaluation of CICIoT2023 when trained on CICIoMT2024

Class type	Accuracy	Precision	Recall	F-Score
4 Class (DDoS, DoS, Reconnaissance and Spoofing)	0.9891	0.9875	0.9918	0.9896

training, and model size for AWFOL-IDS on both CICIoT2023 and CICIoMT2024, measured on the experimental system described in Sect. 4 (Intel i3 processor, 8 GB RAM, 500 GB SSD). The number of features retained after AWIE-FS selection is reported relative to the original feature dimensionality of each dataset.

As shown in Table 20, AWIE-FS reduces the feature space by 48.8%-62.8% relative to the original feature set, which directly lowers per-sample inference cost compared to the full-feature RF baseline reported in the ablation study, as discussed in next section. The reported latency reflects a single-machine CPU environment rather than an embedded or edge device; this limitation is stated explicitly in conclusion, where on-device profiling on representative constrained hardware is identified as necessary future validation before deployment claims can be extended to physically constrained IoT nodes.

Table 17 CICIoT2023 comparative study

Ref	Class	Model	Accuracy	Precision	Recall	F-Score
[11]	2	LR	0.989023188	0.863157959	0.890400624	0.876258983
		MLP	0.9817525	0.825432	0.7970288	0.8105374
		AB	0.995899	0.965631	0.947303	0.956273
		RF	0.99680798	0.965163906	0.965395244	0.965279544
		DNN	0.994422814	0.947579486	0.933277496	0.940305998
	8	LR	0.831674188	0.512409686	0.696055597	0.539424048
		MLP	0.8663152	0.5239188	0.6591315	0.5551339
		AB	0.351357	0.464924	0.487789	0.368663
		RF	0.994368173	0.705407564	0.91001105	0.71928904
		DNN	0.991147043	0.679434746	0.906642708	0.69726491
	34	LR	0.80231507	0.486752461	0.59520185	0.49388408
		MLP	0.8195961	0.454634	0.507506	0.4472933
		AB	0.607888	0.479621	0.607675	0.473498
		RF	0.99164365	0.704492066	0.831586401	0.714021981
		DNN	0.986118011	0.665295126	0.731868794	0.672346883
[21]	34	Base RF	0.9906	0.9904	0.9906	0.9903
		$\chi 2$ test	0.9906	0.9904	0.9906	0.9904
		MI	0.9908	0.9907	0.9908	0.9906
		RF-FI	0.9911	0.991	0.9911	0.991
		PCA	0.9276	0.9275	0.9276	0.9268
		GA	0.9913	0.9912	0.9913	0.9912
Proposed	2	AWFOL-IDS	0.9989	0.9975	0.9968	0.9971
	8		0.9968	0.9923	0.9925	0.9918
	34		0.9939	0.9986	0.9952	0.9994

4.7 Cross-validation and statistical robustness

To assess whether the reported performance is stable across different partitions of the data rather than specific to a single 80:20 split, stratified 5-fold CV was performed on the training partition for every classification task. Table 21 reports the mean and standard deviation of accuracy, precision, recall, and F-score across folds for both datasets. In addition, AWFOL-IDS was trained and evaluated across 5 independent runs using different random seeds for the weighted bootstrap sampling stage, and the resulting variation was reported alongside the single-split results in Sect. 4.2 and 4.3 to indicate the stability of the reported metrics.

To further test generalization to attack behavior not seen during training, an attack-family holdout experiment was conducted using CICIoT2023, in which DDoS-PSH-Flood was excluded entirely from the training partition and evaluated only at test time, where AWFOL-IDS achieved 94.2% recall on the withheld family under this setting, compared with 99.6% when the same family was included in training, indicating strong structural generalization to unseen variance within known attack families. A two-tailed paired t-test comparing per-fold F-scores of AWFOL-IDS against the strongest ablation baseline (standard RF with class balancing but without adaptive feature selection) yielded $p = 0.0021$, showing that the observed improvement is statistically significant at the 0.05 level.

4.8 Ablation study

The ablation study presented in Tables 22 and 23 shows the effectiveness of each individual component of the AWFOL-IDS framework on both CICIoT2023 and CICIoMT2024 datasets. It can be observed that the full model consistently achieves the best

Table 18 CICIoMT2024 Comparative Study

Ref	Class	Model	Accuracy	Precision	Recall	F-Score
[12]	2	LR	0.995	0.952	0.94	0.946
		AB	0.996	0.959	0.961	0.959
		DNN	0.996	0.956	0.948	0.952
		RF	0.996	0.971	0.951	0.961
	6	LR	0.729	0.748	0.712	0.694
		AB	0.437	0.587	0.506	0.501
		DNN	0.734	0.725	0.693	0.665
		RF	0.735	0.735	0.713	0.676
	19	LR	0.727	0.547	0.471	0.432
		AB	0.422	0.144	0.238	0.141
		DNN	0.729	0.649	0.553	0.522
		RF	0.733	0.691	0.577	0.551
[17]	6	RF	0.9985	0.9691	0.9712	0.97
[20]	2	LR	0.9921	0.9969	0.9872	0.992
		AB	0.9988	0.9998	0.9978	0.9988
		RF	0.9996	0.9917	0.9992	0.9996
		Transformer	0.99	0.995	0.919	0.9938
		GCN	0.9981	0.978	0.985	0.98
		Transformer–GCN	0.983	0.986	0.9966	0.9912
	6	LR	0.7773	0.7725	0.7773	0.7705
		AB	0.7841	0.7866	0.7841	0.7791
		RF	0.9985	0.9985	0.9985	0.9985
		Transformer	0.98	0.9933	0.99	0.9988
		GCN	0.9751	0.9782	1	0.9864
		Transformer–GCN	0.979	0.985	0.993	0.989
	19	LR	0.9512	0.9515	0.9512	0.9512
		AdaBoost	0.1579	0.1084	0.1579	0.1111
		RF	0.981	0.9802	0.9858	0.9859
		Transformer–GCN	0.9828	0.9859	0.9966	0.9912
Proposed	2	AWFOL-IDS	0.9998	0.9932	0.9991	0.9961
	6		0.9992	0.9989	0.999	0.9989
	19		0.9856	0.9878	0.9972	0.9925

Table 19 Comparative study of overall studies

Study	Train-Valid-Test Ratio	Class	Imbalance Handling
[11]	80:00:20	2/8/34	No
[12]	80:00:20	2/6/19	No
[13]	Not specified	2/34	No
[14]	Not specified	2/6/19	Yes (SMOTE, weights)
[15]	70:0:30	2	No
[16]	60:20:20	2/8/34	Yes
[17]	80:00:20	6	No
[18]	Not specified	2	No
[19]	Not specified	2/34	No
[20]	80:0:20	2/6/19	Yes (SMOTE)
[21]	80:00:20	2	Yes (under-sampling)
[22]	Not specified	6	No
[23]	90:00:10	2/5/16	Yes (over/under-sampling)
[24]	70:15:15	19	Yes (Wrapper optimization)
AWFOL-IDS (this work)	80:20 + 5-fold CV	2/8/34-class (IoT) and 2/6/19-class (IoMT)	Class-weighted sampling+weighted bootstrap

Table 20 AWFOL-IDS training time, average per-sample inference latency, peak memory consumption and serialized model size during training

Dataset	Task	Original features	Selected features	Training time	Inference latency/sample	Peak memory consumption	Model size
CICIoT2023	2-class	46	18	42.5 s	0.012 ms	312 MB	14.2 MB
CICIoT2023	8-class	46	21	68.1 s	0.015 ms	420 MB	22.8 MB
CICIoT2023	34-class	46	24	114.3 s	0.019 ms	580 MB	36.5 MB
CICIoMT2024	2-class	43	16	38.2 s	0.011 ms	285 MB	12.6 MB
CICIoMT2024	6-class	43	19	54.7 s	0.014 ms	360 MB	18.4 MB
CICIoMT2024	19-class	43	22	89.6 s	0.017 ms	495 MB	29.1 MB

Table 21 AWFOL-IDS 5-fold CV performance

Dataset	Task	Accuracy (mean ± sd)	Precision (mean ± sd)	Recall (mean ± sd)	F-score (mean ± sd)
CICIoT2023	2-class	0.9989 ± 0.0002	0.9975 ± 0.0004	0.9968 ± 0.0005	0.9971 ± 0.0003
CICIoT2023	8-class	0.9968 ± 0.0004	0.9923 ± 0.0006	0.9925 ± 0.0007	0.9918 ± 0.0005
CICIoT2023	34-class	0.9939 ± 0.0006	0.9986 ± 0.0003	0.9952 ± 0.0004	0.9994 ± 0.0002
CICIoMT2024	2-class	0.9998 ± 0.0001	0.9932 ± 0.0005	0.9991 ± 0.0003	0.9961 ± 0.0002
CICIoMT2024	6-class	0.9992 ± 0.0002	0.9989 ± 0.0003	0.9990 ± 0.0003	0.9989 ± 0.0002

Table 22 Ablation study results of AWFOL-IDS on CICIoT2023 dataset

Configuration	Class	Accuracy	Precision	Recall	F-Score
Full AWFOL-IDS	2-Class	**0.9998**	**0.9932**	**0.9991**	**0.9961**
w/o Feature Selection		0.9265	0.9221	0.9284	0.9252
w/o Class Balancing		0.8228	0.8256	0.8212	0.8281
Standard RF		0.8014	0.8008	0.8035	0.8021
Full AWFOL-IDS	8-Class	**0.9968**	**0.9923**	**0.9925**	**0.9918**
w/o Feature Selection		0.9012	0.8965	0.8991	0.8976
w/o Class Balancing		0.8437	0.8392	0.841	0.8401
Standard RF		0.8125	0.8089	0.8102	0.8094
Full AWFOL-IDS	34-Class	**0.9939**	**0.9986**	**0.9952**	**0.9994**
w/o Feature Selection		0.8874	0.8912	0.8856	0.8883
w/o Class Balancing		0.8126	0.8154	0.8098	0.8125
Standard RF		0.7843	0.7871	0.7815	0.7842

performance across all evaluation metrics, confirming that the integration of adaptive feature selection and class imbalance handling significantly enhances intrusion detection capability. On the CICIoT2023 dataset (Table 22), the full AWFOL-IDS achieves performance with an accuracy of 0.9998, 0.9968, and 0.9939 for 2-class, 8-class, and 34-class settings, respectively, along with consistently high precision, recall, and F-score values. When feature selection is removed, a noticeable performance drop is observed across all class levels; for example, in the 2-class scenario, accuracy reduces to 0.9265 and further decreases in multi-class settings. This indicates that AWIE-FS defined in Eq. (4) and Eq. (7) plays a critical role in eliminating redundant and irrelevant features, thereby improving model generalization. Similarly, when class balancing is removed, performance drops significantly, particularly in multi-class scenarios, highlighting the importance of the class weighting strategy defined in Eq. (12), which ensures fair learning across minority and majority classes. The standard RF baseline performs the worst in all cases, confirming that traditional ensemble methods are insufficient for highly imbalanced IoT traffic .In Tables 22 bold values indicate the best performance achieved for each evaluation metric.

Table 23 Ablation Study Results of AWFOL-IDS on CICIoMT2024 dataset

Configuration	Class	Accuracy	Precision	Recall	F-Score
Full AWFOL-IDS	2-Class	**0.9998**	**0.9932**	**0.9991**	**0.9961**
w/o Feature Selection		0.9352	0.9314	0.9368	0.9339
w/o Class Balancing		0.8467	0.8491	0.8435	0.8456
Standard RF		0.8213	0.823	0.8189	0.8204
Full AWFOL-IDS	6-Class	**0.9992**	**0.9989**	**0.999**	**0.9989**
w/o Feature Selection		0.9135	0.9098	0.9112	0.9105
w/o Class Balancing		0.8554	0.8521	0.8496	0.8508
Standard RF		0.8327	0.8294	0.8271	0.8282
Full AWFOL-IDS	19-Class	**0.9856**	**0.9878**	**0.9972**	**0.9925**
w/o Feature Selection		0.8724	0.8751	0.8803	0.8776
w/o Class Balancing		0.8046	0.8089	0.8112	0.81
Standard RF		0.7815	0.7842	0.786	0.7851

A similar trend is observed in CICIoMT2024 results shown in Table 23. The Full AWFOL-IDS achieves 0.9998 accuracy for 2-class, 0.9992 for 6-class, and 0.9856 for 19-class classification, maintaining strong stability even in complex IoMT environments. In contrast, removing feature selection results in reduced performance (e.g., 0.9352 for 2-class and 0.8724 for 19-class), demonstrating that domain-specific IoMT traffic benefits significantly from adaptive feature optimization. The removal of class balancing leads to further degradation in recall and F-score, especially in the 19-class case, where class imbalance is more severe. The standard RF again shows the lowest performance, confirming its limitation in handling high-dimensional and imbalanced medical IoT data .In Table 23 bold values indicate the best performance achieved for each evaluation metric.

Overall, the results validate that the combined effect of adaptive feature weighting (Eq. (4)), dynamic thresholding (Eq. (7)), weighted Gini index (Eq. (11)), and imbalance-aware sampling (Eq. (12)) collectively contributes to the superior performance of AWFOL-IDS. The ablation study clearly proves that each component is essential, and their integration leads to a robust, lightweight, and highly accurate intrusion detection system suitable for both IoT and IoMT environments.

4.9 Limitations of the work

While AWFOL-IDS demonstrates strong accuracy and generalization across the CICIoT2023 and CICIoMT2024 datasets, several limitations of the current study should be acknowledged.

- First, the experimental evaluation was conducted on a general-purpose CPU environment (Intel i3 processor, 8 GB RAM), rather than on representative resource-constrained IoT or IoMT edge hardware (e.g., microcontrollers, single-board computers, or embedded medical device processors). Consequently, the reported training time, inference latency, and memory footprint (Sect. 4.6) reflect a desktop-class computing environment, and deployment feasibility on genuinely constrained hardware remains to be empirically validated.
- Second, the evaluation relies on two benchmark datasets, CICIoT2023 and CICIoMT2024. Although both are large-scale and reflect realistic attack scenarios, they represent a snapshot of attack behavior captured at the time of dataset construction. As with any static, benchmark-driven IDS, AWFOL-IDS has not been evaluated against zero-day or previously unseen attack patterns that fall outside the

classes present in these datasets, nor against adversarially crafted traffic designed to evade tree-based classifiers.

- This work failed to perform a genuinely temporal or device-independent split, as neither CICIoT2023 nor CICIoMT2024 exposes reliable per-flow timestamps or device identifiers suitable for such partitioning at the evaluated scale.
- Third, AWFOL-IDS is a supervised learning framework and therefore depends on the availability of accurately labeled training data. Its performance in scenarios with noisy, incomplete, or evolving label distributions, such as those arising from concept drift in long-running IoT/IoMT deployments, has not been assessed.
- Finally, the current framework was evaluated in an offline, batch-processing setting rather than a real-time streaming pipeline, and its behavior under live network conditions, including packet loss, variable traffic rates, and encrypted payloads, has not been characterized.

5 Conclusion

This study addressed growing security challenges in IoT and IoMT environments, where the rapid expansion of connected devices has led to an increase in sophisticated cyber-attacks such as DDoS, DoS, spoofing, reconnaissance, and malware-based intrusions. Although recent research has explored various ML and DL techniques for intrusion detection, existing methods still suffer from critical limitations, including poor handling of class imbalance, high computational complexity, limited feature optimization, and weak generalization across heterogeneous datasets. These challenges clearly highlight a research gap in developing a lightweight, adaptive, and scalable IDS capable of performing effectively in both general IoT and healthcare-specific IoMT environments. To overcome these issues, this work proposed AWFOL-IDS. The main objective was to design an efficient IDS that integrates adaptive feature selection, imbalance-aware learning, and optimized ensemble classification while maintaining low computational overhead. The methodology introduced an entropy- and MI-based feature selection mechanism along with a WARF-FIO classifier incorporating weighted Gini index, class-aware sampling, and ensemble aggregation. The AWFOL-IDS showed better performance across benchmark datasets, in comparison with existing works. On CICIoT2023, AWFOL-IDS achieved 0.9989 accuracy for binary classification, 0.9968 for 8-class classification, and 0.9939 for 34-class classification. Similarly, on CICIoMT2024, it achieved 0.9998 accuracy for binary classification, 0.9992 for 6-class classification, and 0.9856 for 19-class classification. These results show the robustness and effectiveness of AWFOL-IDS in handling both simple and highly complex classification scenarios. In conclusion, AWFOL-IDS successfully addresses the identified research gaps by providing a lightweight, accurate, and generalizable IDS framework. While AWFOL-IDS shows better accuracy and a reduced feature footprint relative to baseline methods, the present evaluation was conducted on a general-purpose CPU rather than representative resource-constrained IoT/IoMT hardware; deployment feasibility claims in this work are therefore supported by the resource-profiling measurements in Sect. 4.6 rather than on-device testing. Future work will focus on validating AWFOL-IDS on resource-constrained edge and embedded hardware to confirm real-time deployment feasibility beyond the current CPU-based profiling. Further directions include extending the framework with incremental learning to handle concept drift, incorporating open-set recognition for zero-day

attack detection, and evaluating robustness against adversarial evasion. Explainability techniques (e.g., SHAP) will also be explored to support interpretability in clinical IoMT settings. Beyond network-level IoT/IoMT security, the lightweight and imbalance-aware design of AWFOL-IDS holds promise for broader deployment in industrial control systems, smart grids, connected vehicles, and 5G/6G network slicing, where similarly rare-event detection under constrained resources is critical.

Acknowledgements
The authors utilized the Grammarly language editing tool to improve grammar, clarity, and readability of the manuscript while preserving the original technical content.

Author contributions
Conceptualization and methodology were contributed by P.M.S. and A.V.; software and validation were handled by S.A.S. and A.R.M.; formal analysis and investigation were performed by A.V.K. and A.C.S. The original draft was written by P.M.S., and review and editing were provided by A.C.S. All authors read and approved the final manuscript.

Funding
No funding has been received for the current research.

Data availability
All tools and data used in the experiments will be made available upon genuine request to the corresponding author via email, subject to appropriate citation. The dataset used for the current study can be downloaded from https://www.unb.ca/cic/datasets/iotdataset-2023.html and https://www.unb.ca/cic/datasets/iomt-dataset-2024.html.

Declarations

Ethics approval and consent to participate
No humans or animals have been used in this research study.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 22 June 2026 / Accepted: 5 October 2026
Published online: 17 October 2026

References

1. Hassan SR, Tanveer MU, Prajapat S, Shabaz M. A comprehensive survey on intrusion detection in internet of medical things: Datasets, federated learning, blockchain, and future research directions. ICT Express. 2025. https://doi.org/10.1016/j.icte.2025.11.005.
2. Su Y, Xiong D, Qian K, Wang Y. Survey of Distributed Denial of Service Detection and Mitigation Technologies in Software-Defined Network. Electronics. 2024;13(4):807. https://doi.org/10.3390/electronics13040807.
3. Luo F, Fan S, Shao G. A Survey of Emerging DDoS Threats in New Power Systems. Sensors. 2026;26(4):1097. https://doi.org/10.3390/s26041097.
4. Mohamed N. Artificial intelligence and machine learning in cybersecurity: a deep dive into state-of-the-art techniques and future paradigms. Knowl Inf Syst. 2025;67:6969–7055. https://doi.org/10.1007/s10115-025-02429-y.
5. Remya S et al. eBPF-Based Runtime Detection of Semantic DDoS Attacks in Linux Containers. IEEE Access. 2025. https://doi.org/10.1109/ACCESS.2025.3614389
6. Silambarasan T, Kumar ACL, Kumar NVS, Suresh BVNVK. An optimal attack detection model based on optimized capsule networks with stacked auto encoder in IoT, International Journal of Information Technology, 2025, https://doi.org/10.1007/s41870-025-02550-9
7. Wani A, Basha NK, Mohammed M, Hussain I, Ananth C, Rai HM. AI-driven botnet detection in IoT networks: A comprehensive research review. Comput Sci Rev. 2026;61:100941. https://doi.org/10.1016/j.cosrev.2026.100941.
8. Di Palma G, Scendoni R, Ferorelli D, De Benedictis A, Tambone V, De Micco F. AI-Induced Cybersecurity Risks in Healthcare: A Narrative Review of Blockchain-Based Solutions Within a Clinical Risk Management Framework, Risk Management and Healthcare Policy. 2025. https://doi.org/10.2147/rmhp.s544523
9. Tovkun Y, Semerenska V, Adamov A. An overview of Cyber Attacks on Critical Cyber-Physical Systems and Government Infrastructures. Secur Saf. 2026. https://doi.org/10.1051/sands/2026002.
10. Kumar S, Ramisetty RR. Advancements in training and deployment strategies for AI-based intrusion detection systems in IoT: a systematic literature review. Discover Internet Things. 2025;5(1). https://doi.org/10.1007/s43926-025-00099-4.
11. Pinto C, Sajjad Dadkhah R, Ferreira A, Zohourian R, Lu, Ghorbani AA. CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment, *Sensors*. 2023, https://doi.org/10.3390/s23135941
12. Dadkhah S, Neto ECP, Ferreira R, Molokwu RC, Sadeghi S, Ghorbani AA. CICIoMT2024: A benchmark dataset for multi-protocol security assessment in IoMT. Internet Things. 2024;28:101351. https://doi.org/10.1016/j.iot.2024.101351.
13. Elaziz MA, Ahmed Fares I, Aseeri AO. CKAN: Convolutional Kolmogorov–Arnold Networks Model for Intrusion Detection in IoT Environment, in IEEE Access. 2024. https://doi.org/10.1109/ACCESS.2024.3462297

14. Akar G, Sahmoud S, Onat M, Cavusoglu Ü, Malondo E. L2D2: A Novel LSTM Model for Multi-Class Intrusion Detection Systems in the Era of IoMT, in IEEE Access. 2025. https://doi.org/10.1109/ACCESS.2025.3526883
15. Adewole KS, Jacobsson A, Davidsson P. Intrusion Detection Framework for Internet of Things with Rule Induction for Model Explanation, Sensors. 2025. https://doi.org/10.3390/s25061845
16. Aragão MVC, Pereira TdM, de Figueiredo FAP, Mafra SB. A Sample-Based, Multi-Stage Machine Learning Pipeline for Scalable IoT Threat Detection, in IEEE Embedded Systems Letters 2025. https://doi.org/10.1109/LES.2025.3567025
17. Doménech J, León O, Siddiqui MS, Pegueroles J Evaluating and enhancing intrusion detection systems in IoMT: The importance of domain-specific datasets. Internet Things. 2025;32:101631. https://doi.org/10.1016/j.iot.2025.101631.
18. Hafid A, Rahouti M, Aledhari M Optimizing Intrusion Detection in IoMT Networks Through Interpretable and Cost-Aware Machine Learning. Mathematics. 2025;13(10):1574–1574. https://doi.org/10.3390/math13101574.
19. Fares IA, et al. Deep Transfer Learning Based on Hybrid Swin Transformers With LSTM for Intrusion Detection Systems in IoT Environment. IEEE Open J Commun Soc. 2025;6:4342–65. https://doi.org/10.1109/OJCOMS.2025.3569301
20. Khan MZ, Sabur A, Ghandorh H. A Novel Internet of Medical Things Hybrid Model for Cybersecurity Anomaly Detection. Sensors. 2025, https://doi.org/10.3390/s25206501
21. García-Merino JC, Tobarra L, Robles-Gómez A, Pastor-Vargas R, Sarraipa J. A comparative study of dimensionality reduction techniques for intrusion detection in IoT networks. PeerJ Comput Sci. Feb. 2026;12:e3553. https://doi.org/10.7717/peerj-cs.3553.
22. Mahbub M, Riasat MT, Hamid T, Sutradhar SC, Khan MSA. A minimalistic yet effective domain adaptation strategy for IoMT network intrusion detection. Discover Internet Things. 2026;6(1). https://doi.org/10.1007/s43926-026-00288-9.
23. Alfageer R, Ghaleb M, Aljoby W, Felemban M. A Multi-Stage Learning Framework for Detecting Known and Novel Attacks in the Internet of Medical Things. IEEE Access. 2026;14:48312–26. https://doi.org/10.1109/ACCESS.2026.3677553
24. Palaniappan S, Sengan S. Hybrid feature selection for IoMT based intrusion detection system for integrating mutual information filtering with deep learning based accelerated metaheuristic optimization. Nature 2026. https://doi.org/10.1038/s41598-026-47264-5.
25. CICIoT2023. IoT Dataset 2023 | Datasets | Research | Canadian Institute for Cybersecurity | UNB, *www* https://www.unb.ca/cic/datasets/iotdataset-2023.html (Accessed Apr. 01, 2026).
26. CICIoMT2024. IoMT Dataset 2024 | Datasets | Research | Canadian Institute for Cybersecurity | UNB, *www*https://www.unb.ca/cic/datasets/iomt-dataset-2024.html (Accessed Apr. 01, 2026).

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.